

Heat Geometry from a Universal Scalar Process

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Abstract

We prove that the law of one smooth scalar process in a finite Wiener chaos completely determines finite families of symmetric tensors on arbitrary real separable Hilbert spaces, up to simultaneous orthogonal equivalence. Gaussian graph characters recover all contractions, while an intrinsic trace class Gram operator reduces the problem to finite dimensional invariant theory. Applied at a single fixed positive time to the canonical quadratic and quartic heat packet, this principle reconstructs the heat generator and forces every reconstructed unitary to be spatial. Consequently, one universal scalar process determines closed Riemannian manifolds, compact RCD spaces of finite dimension, and Euclidean bundles with metric connection and self adjoint potential. The heat packet functor is fully faithful, and its tensor automorphisms are precisely the geometric ones.

Keywords. Replica–Tannaka reconstruction; Gaussian canvas; Wiener chaos; orthogonal tensor invariants; heat kernel; inverse spectral geometry; RCD spaces; connection Laplacian.

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1 Introduction

The spectrum of the Laplace operator is a powerful but incomplete geometric invariant. Kac’s question, Milnor’s flat tori, Vignéras’ arithmetic examples, Sunada’s construction, and the planar examples of Gordon–Webb–Wolpert established the fundamental gap between unitary spectral equivalence and geometric equivalence [Kac66, Mil64, Vig80, Sun85, GWW92]. The missing datum is spatial: a unitary intertwiner of Laplacians need not preserve positivity or pointwise multiplication. Once an intertwiner is an order isomorphism, diffusion recovers the manifold and, more generally, the intrinsic geometry of a Dirichlet space [ABtE12, LSW18].

This paper reconstructs that spatial structure from a single scalar probabilistic law. The observation architecture is fixed before the unknown object is chosen. It consists of one stationary Gaussian canvas on the real line with covariance

$$K(u, v) = e^{-(u-v)^2}. \quad (1)$$

Every finite symmetric tensor packet on every real separable Hilbert space is placed on this same canvas and becomes one smooth scalar finite-chaos process. Its law determines the entire tensor packet up to one common orthogonal transport. When the packet is the canonical quadratic–quartic heat packet, that transport is automatically a geometric isometry–gauge arrow.

The result has three levels. The first is an infinite-dimensional orthogonal reconstruction theorem. The second identifies the canonical geometric packet and proves its full faithfulness. The third organizes its finite spectral cuts into a positive orthogonal Tannakian inverse system.

1.1 Universal scalar Replica–Tannaka reconstruction

Let H be a real separable Hilbert space and let

$$\mathbf{f} = (f_0, f_1, \dots, f_m), \quad f_0 \in \mathbb{R}, \quad f_q \in H^{\odot q}, \quad (2)$$

be a finite symmetric tensor packet. Over H take a jointly Gaussian family $(\mathbb{W}_u)_{u \in \mathbb{R}}$ satisfying

$$\mathbb{E}[\mathbb{W}_u(h)\mathbb{W}_v(k)] = e^{-(u-v)^2} \langle h, k \rangle_H. \quad (3)$$

Define

$$\mathcal{Z}_{\mathbf{f}}(u) = f_0 + \sum_{q=1}^m I_q^{\mathbb{W}_u}(f_q), \quad u \in \mathbb{R}. \quad (4)$$

The process has a canonical almost surely smooth realization. Write $S(\mathbf{f})$ for the smallest closed subspace $S \subset H$ with $f_q \in S^{\odot q}$ for every $q \geq 1$.

Theorem 1.1 (Universal scalar Replica–Tannaka reconstruction). *Let $\mathbf{f}$ on H and $\mathbf{f}'$ on H' be finite symmetric tensor packets. Complete the shorter packet by zero tensors. Then*

$$\text{Law}_{C(\mathbb{R})}(\mathcal{Z}_{\mathbf{f}}) = \text{Law}_{C(\mathbb{R})}(\mathcal{Z}_{\mathbf{f}'}) \iff \begin{cases} f_0 = f'_0, \\ \exists U : S(\mathbf{f}) \rightarrow S(\mathbf{f}') \text{ orthogonal and onto,} \\ f'_q = U^{\odot q} f_q \text{ for every } q \geq 1. \end{cases} \quad (5)$$

The same orthogonal map transports every tensor grade.

The proof exposes the complete invariant rather than invoking an abstract encoding. Correlated copies of one Wiener polynomial produce mixed moments whose coefficients are closed loop-free tensor networks. On the fixed canvas, a multigraph with adjacency multiplicities N contributes the Gaussian graph character

$$\chi_N(x) = \exp \left(- \sum_{a < b} N_{ab} (x_a - x_b)^2 \right). \quad (6)$$

The edge forms $(x_a - x_b)^2$ form a basis of $\text{Sym}^2(\mathbf{1}^\perp)^*$, hence the characters χ_N are linearly independent. Their Gaussian Gram matrix has the closed form

$$G_{N,N'} = \det_{\mathbf{1}^\perp} (I + L_N + L_{N'})^{-1/2}, \quad (7)$$

and G^{-1} recovers every network coefficient. This is a genuine compression: the configuration space has dimension $r - 1$, while the unrestricted correlation register has $r(r - 1)/2$ coordinates.

Loop-free networks do not yet contain self-contractions. The decisive infinite-dimensional device is the intrinsic trace-class support Gram

$$D_{\mathbf{f}} = \sum_{q=1}^m L_{f_q} L_{f_q}^*. \quad (8)$$

Its positive spectral cuts are finite-dimensional and recover all missing contractions by trace-norm polynomial approximation. The orthogonal first fundamental theorem then separates each finite cut, and the marker $P_\delta D_{\mathbf{f}} P_\delta$ makes the finite transports coherent. Compactness of the inverse system yields one orthogonal map on the dense union of the cuts. This combines classical Wick–Isserlis diagrammatics and orthogonal invariant theory [Iss18, Jan97, PT11, Wey46, Bra37, Sch08, You25] with a new scalar decoder and a new separable-Hilbert inverse-limit argument.

The construction is functorial. Orthogonal direct sums become convolution of process laws, orthogonal transports leave the law unchanged, and Hilbert tensor convergence gives quantitative convergence in the compact-open topology. The covariance (1) is itself canonical in the replica-stable stationary class: stationarity, nondegeneracy at the origin, and the identity $K(\sqrt{n}u, \sqrt{n}v) = K(u, v)^n$ force a Gaussian covariance up to length scale. At the feature-space level, tensoring replicas is exactly dilation of the canvas coordinate.

1.2 The quadratic–quartic master packet

Let

$$\mathfrak{X} = (M, g, E, \nabla^E, V_E), \quad \mathcal{L}_{\mathfrak{X}} = (\nabla^E)^* \nabla^E + V_E, \quad (9)$$

where M is closed and connected, $E \rightarrow M$ is a positive-rank real Euclidean bundle, ∇^E is metric, and V_E is smooth and self-adjoint. At a fixed $t > 0$, heat evaluation defines tensors

$$Q_{\mathfrak{X},t} \in L^2(M, E)^{\odot 2}, \quad C_{\mathfrak{X},t} \in L^2(M, E)^{\odot 4}. \quad (10)$$

They satisfy the two identities

$$\text{Flat}(Q_{\mathfrak{X},t}) = e^{-t\mathcal{L}_{\mathfrak{X}}}, \quad (11)$$

$$\langle C_{\mathfrak{X},t}, s^{\otimes 4} \rangle = \|e^{-t\mathcal{L}_{\mathfrak{X}}/2} s\|_{L^4(M,E)}^4. \quad (12)$$

The quadratic tensor recovers the complete generator by the unbounded logarithm. The quartic tensor supplies the first non-Hilbertian radial norm and turns the same reconstructed L^2 -unitary into an L^4 isometry. Vector-valued Banach–Lamperti rigidity makes that unitary spatial; elliptic regularity and comparison of the principal, first-order, and zeroth-order symbols recover respectively the metric, connection, and potential.

Proposition 1.2 (Minimality of the positive radial packet). *For every nonzero finite-dimensional Euclidean space V ,*

$$\mathbb{R}[V]^{O(V)} = \mathbb{R}[|v|^2]. \quad (13)$$

Hence odd radial tensor degrees vanish, degree two records only the Hilbert norm, and degree four is the first homogeneous radial invariant that supplies an L^p geometry with $p \neq 2$. The packet (Q_t, C_t) is therefore the minimal positive radial heat packet that is intrinsic under arbitrary orthogonal changes of fiber frame and works uniformly in every finite rank.

In the scalar line-bundle sector there is also a native cubic multiplication tensor. Its preservation together with Q_t forces a unital algebra isomorphism on smooth functions. The stronger status of (Q_t, C_t) is its positivity and its uniform orthogonal-radial meaning for scalar functions, singular diffusions, and bundles of arbitrary rank.

Theorem 1.3 (Full reconstruction and arrow-level rigidity). *Fix $t > 0$ and let $\mathfrak{X}, \mathfrak{Y}$ be smooth heat geometries of the form (9). Form the scalar canvas processes*

$$\mathcal{Z}_{\mathfrak{X},t}(u) = I_2^{\mathbb{W}_u}(Q_{\mathfrak{X},t}) + I_4^{\mathbb{W}_u}(C_{\mathfrak{X},t}). \quad (14)$$

Then the following are equivalent:

- (i) *the $C(\mathbb{R})$ -valued laws of $\mathcal{Z}_{\mathfrak{X},t}$ and $\mathcal{Z}_{\mathfrak{Y},t}$ agree;*
- (ii) *one orthogonal isomorphism transports both Q_t and C_t ;*
- (iii) *there exist a Riemannian isometry of the bases and a smooth fiberwise orthogonal bundle isomorphism transporting the connection and potential.*

Every tensor-transporting orthogonal isomorphism is induced by one unique geometric arrow. Thus the master heat-packet functor is fully faithful, and

$$\mathrm{Aut}(\mathfrak{X}) \simeq \mathrm{Stab}_{O(L^2(M,E))}(Q_{\mathfrak{X},t}, C_{\mathfrak{X},t}). \quad (15)$$

For the trivial real line bundle, this proves that one universal scalar process determines the unpointed Riemannian isometry class of a closed connected manifold. For general Euclidean bundles it reconstructs the bundle topology, rank, connection, curvature, holonomy, and smooth self-adjoint potential up to the sharp isometry–orthogonal-gauge action. No dimension, rank, or spectral simplicity is supplied in advance; these are outputs of the reconstruction.

1.3 Singular heat geometry

The same mechanism is intrinsic to symmetric diffusion. If a finite Dirichlet space has a heat kernel and a dense heat range in L^4 , the quadratic–quartic packet reconstructs the measure algebra and the entire Dirichlet form. Compact finite-dimensional RCD spaces satisfy these hypotheses through heat-kernel bounds, analytic semigroup theory, and the Sobolev-to-Lipschitz identification of the intrinsic metric [AGS14a, AGS14b, EKS15, JLZ16, Gig18].

Theorem 1.4 (One scalar process reconstructs compact RCD spaces). *Fix $t > 0$. Let (X_i, d_i, μ_i) , $i = 1, 2$, be compact connected $\mathrm{RCD}(K_i, N_i)$ spaces with full support and finite synthetic dimension. The scalar canvas processes formed from their Cheeger heat flows at time t have the same law if and only if there is a measure-preserving isometry*

$$(X_2, d_2, \mu_2) \simeq (X_1, d_1, \mu_1). \quad (16)$$

Thus the same scalar architecture reconstructs smooth manifolds, singular metric-measure spaces, and orthogonal gauge geometry. The common principle is not a coordinate formula; it is the recovery of spatial multiplication from one common orthogonal transport of the quadratic and quartic heat tensors.

1.4 Positive orthogonal Tannaka duality

Every regular spectral cut of the support Gram is a finite-dimensional orthogonal tensor object generated by the compressed marker and the compressed tensors of degrees two and four. Its closed diagrams define a positive character, and the corresponding tensor automorphism group is the compact stabilizer of that finite cut. The cut characters form a cofinal inverse system. Their projective-limit automorphism group is the full Hilbert-space stabilizer, which is the geometric automorphism group by Theorem 1.3. This gives a concrete positive orthogonal Tannaka duality, in the finite-cut sense natural to a separable Hilbert object, and links the scalar law, the closed-network character, and the geometric groupoid. The categorical background is provided by classical Tannaka theory and compact-group reconstruction [Saa72, Del02, DR89, EGNO15].

The canonical invariant is mixed of degrees two and four, but its orbit can be carried by any fixed homogeneous chaos. Anchor padding gives faithful degree- q presentations for $q \geq 4$; a minimal positive factorization of the $2 + 2$ flattening of C_t gives a fully faithful cubic dilation. A typed version of the universal canvas theorem then recovers each marked homogeneous packet from one process lying entirely in the pure q th Wiener chaos.

Theorem 1.5 (Homogeneous carriers of the master packet). *For every integer $q \geq 3$, there is an orthogonally natural finite marked family of degree- q tensors whose common orthogonal orbit is equivalent to the orbit of the master packet (Q_t, C_t) . The law of one typed scalar process in the pure q th Wiener chaos determines this family and therefore the complete heat geometry. The cubic construction is fully faithful; for even $q \geq 4$ the only extra symmetry is the canonical reflection of the auxiliary anchor line.*

1.5 Relation to previous work

The finite-dimensional Gaussian orbit problem is solved in [Ass26a]: a prescribed finite family of rational, uniformly nondegenerate correlated Gaussian experiments recovers a polynomial orbit and is bi-Hölder equivalent to orbit distance on coefficient balls. The present theorem crosses two new boundaries. It reconstructs arbitrary separable-Hilbert tensor packets, and it replaces the entire freely chosen covariance register by one fixed scalar process on a one-dimensional canvas. The support-Gram spectral flag is the mechanism that converts loop-free scalar data into complete infinite-dimensional orbit reconstruction.

The minimal graded-support and homogeneous-realization viewpoint is closely related to the Primitive–Fock classification of weak chaos limits in [Ass26b]; recent structural results on limits of Gaussian polynomials provide a complementary asymptotic perspective [HMP24]. Here the problem is exact rather than asymptotic: a fixed process law reconstructs every tensor grade and every transport between geometric realizations.

Classical orthogonal tensor invariants are generated by contractions [Wey46, Bra37, Sch08]; graph-algebra and reflection-positive reconstruction provide a related deterministic language for partition functions [FLS07, LS09]. The scalar replica law differs in that neither a labelled network oracle nor a finite active dimension is given. The graph-character decoder and spectral-cut opening recover both from the law itself.

Several enrichments of ordinary spectral data reconstruct geometry. Order-intertwining diffusion determines smooth and intrinsic Dirichlet geometry [ABtE12, LSW18]; Wick squares of Gaussian free fields yield Riemannian rigidity [Dan22]; eigenfunction triple products and local Weyl data give other complete enhancements [Sch24, WWX26]. The datum here is a single fixed-architecture scalar process law. Its quartic component reconstructs the order structure that those deterministic rigidity theorems require.

For connection Laplacians, local hyperbolic boundary data reconstruct manifolds and compatible connections [KOP18], while Bochner spectral rigidity is known in geometric regimes such as negatively curved low-rank settings [CL25]. The present theorem is global, fixed-time, and arrow-level: it identifies every simultaneous transport of the two heat tensors with a unique isometry–gauge transformation.

1.6 Organization

Section 2 extracts loop-free contractions from scalar replica moments. Section 3 reconstructs the complete tensor orbit in a separable Hilbert space through the support Gram, finite spectral cuts, orthogonal invariant theory, and a coherent inverse limit; it then replaces free correlations by the fixed Gaussian canvas. Section 4 exhibits the information gained by replicas beyond one-point marginals. The smooth manifold, Dirichlet–RCD, and gauge reconstructions are proved in Sections 5 to 7. Section 8 identifies the master packet and its closed networks, while Section 9 proves positive scalar Tannaka duality and full faithfulness. The homogeneous carriers, sharp inverse-stability results, and finite statistical identification theorems are collected in the appendices.

2 Correlated replicas and the scalar register

Throughout the reconstruction part of the paper, H denotes a real separable Hilbert space. An *isonormal Gaussian process* over H is a centered Gaussian linear map

$$W : H \longrightarrow L^2(\Omega) \quad \text{such that} \quad \mathbb{E}[W(h)W(k)] = \langle h, k \rangle_H.$$

For $q \geq 1$, we write $I_q^W(f)$ for the q th multiple Wiener integral of $f \in H^{\odot q}$, with the normalization

$$\mathbb{E}[I_q^W(f)I_p^W(g)] = \mathbf{1}_{\{p=q\}} q! \langle f, g \rangle_{H^{\otimes q}}. \quad (17)$$

Fix $m \geq 1$ and a finite symmetric tensor packet

$$\mathbf{f} = (f_0, f_1, \dots, f_m), \quad f_0 \in \mathbb{R}, \quad f_q \in H^{\odot q}, \quad (18)$$

and set

$$F_{\mathbf{f}}(W) = f_0 + \sum_{q=1}^m I_q^W(f_q). \quad (19)$$

For $r \geq 1$, let

$$\text{Corr}_r^\circ = \left\{ \Sigma \in \mathbb{R}^{r \times r} : \Sigma = \Sigma^\top > 0, \quad \Sigma_{aa} = 1 \ (1 \leq a \leq r) \right\}. \quad (20)$$

Given $\Sigma \in \text{Corr}_r^\circ$, a Σ -correlated isonormal family $(W_1, \dots, W_r)$ over H is defined by

$$\mathbb{E}[W_a(h)W_b(k)] = \Sigma_{ab} \langle h, k \rangle_H. \quad (21)$$

Such a family exists: if $AA^\top = \Sigma$ and $\widetilde{W}_1, \dots, \widetilde{W}_r$ are independent isonormal processes, then $W_a = \sum_j A_{aj} \widetilde{W}_j$ has (21).

Definition 2.1 (Scalar replica signature). *The correlated-replica signature of $\mathbf{f}$ is the indexed family of characteristic functions*

$$\text{Rep}(\mathbf{f}) = (\Phi_{\mathbf{f}}(r, \Sigma, z))_{r \geq 1, \Sigma \in \text{Corr}_r^\circ, z \in \mathbb{R}^r}, \quad \Phi_{\mathbf{f}}(r, \Sigma, z) = \mathbb{E} \exp \left(i \sum_{a=1}^r z_a F_{\mathbf{f}}(W_a) \right). \quad (22)$$

Equality of signatures means pointwise equality for every displayed index. In particular, the datum consists only of joint laws of copies of the one scalar random variable $F_{\mathbf{f}}$; no chaos projection, tensor leg, or open network is supplied to the observer.

The following elementary lemma makes the passage from laws to moments explicit.

Lemma 2.2 (Replica laws determine joint moments). *Let $Z = (Z_1, \dots, Z_s)$ be a finite family contained in the sum of the first m Wiener chaoses over an arbitrary Gaussian Hilbert space. Every polynomial moment of Z is finite and is determined by its joint law. Consequently, (22) determines*

$$M_{\mathbf{f},r}(\Sigma) := \mathbb{E} \prod_{a=1}^r F_{\mathbf{f}}(W_a)$$

for every r and every $\Sigma \in \text{Corr}_r^\circ$.

Proof. Let J_q denote orthogonal projection onto the q th chaos. Nelson's hypercontractive estimate [Jan97] gives, for $p \geq 2$,

$$\|J_q Z_j\|_{L^p} \leq (p-1)^{q/2} \|J_q Z_j\|_{L^2}.$$

Hence every Z_j belongs to every finite L^p , and Hölder's inequality gives all mixed polynomial moments. The characteristic function determines the Borel law. If z^α is a monomial, truncate it to the cube $\max_j |z_j| \leq R$ and let $R \rightarrow \infty$; dominated convergence recovers its expectation from the law. This avoids any unjustified differentiation under the expectation. $\square$

2.1 Extraction of loop-free contractions

Let $N = (N_{ab})_{1 \leq a, b \leq r}$ be a symmetric matrix of nonnegative integers such that

$$N_{aa} = 0, \quad q_a := \sum_{b \neq a} N_{ab} \in \{1, \dots, m\}. \quad (23)$$

At vertex a place the symmetric tensor f_{q_a} and join exactly N_{ab} tensor legs between vertices a and b . Since the tensors are symmetric, the resulting scalar depends only on N ; we denote it by

$$\text{Contr}_N(f_{q_1}, \dots, f_{q_r}). \quad (24)$$

Parallel edges are allowed, whereas an edge from a vertex to itself is not. Put $\Sigma^N = \prod_{a < b} \Sigma_{ab}^{N_{ab}}$.

Proposition 2.3 (Exact scalar coefficient formula). *For every N satisfying (23),*

$$[\Sigma^N] M_{\mathbf{f}, r}(\Sigma) = \frac{\prod_{a=1}^r q_a!}{\prod_{1 \leq a < b \leq r} N_{ab}!} \text{Contr}_N(f_{q_1}, \dots, f_{q_r}). \quad (25)$$

Consequently, $\text{Rep}(\mathbf{f})$ determines every closed loop-free tensor network made of the kernels $f_1, \dots, f_m$, with arbitrary repetitions.

Proof. Expand each factor in $M_{\mathbf{f}, r}$ into homogeneous chaoses. Wick's formula pairs microscopic tensor legs belonging to distinct Wick-ordered vertices. A pairing therefore has a symmetric zero-diagonal adjacency matrix N , and each edge between replicas a and b contributes the factor Σ_{ab} . Thus $M_{\mathbf{f}, r}$ is a polynomial in the $\binom{r}{2}$ off-diagonal entries of Σ .

For the monomial Σ^N , the degree condition at vertex a forces the unique homogeneous kernel f_{q_a} . To count the pairings, first partition the q_a labelled leg positions at vertex a into blocks of sizes $(N_{ab})_{b \neq a}$, then biject the two blocks assigned to each unordered pair $\{a, b\}$. The number of pairings is

$$\prod_a \frac{q_a!}{\prod_{b \neq a} N_{ab}!} \prod_{a < b} N_{ab}! = \frac{\prod_a q_a!}{\prod_{a < b} N_{ab}!}.$$

All these pairings have the same value because every f_q is symmetric, which proves (25).

It remains only to justify coefficient extraction from correlation matrices with fixed diagonal. If $A(z)$ is symmetric with zero diagonal and off-diagonal entries z_{ab} , then

$$\max_{a < b} |z_{ab}| < \frac{1}{r-1} \implies \|A(z)\|_{\text{op}} < 1 \implies I_r + A(z) > 0.$$

Thus Corr_r° contains a genuine open neighbourhood of I_r in all off-diagonal coordinates. A polynomial is determined by its values on that open set, so [Lemma 2.2](#) determines every coefficient in (25). The constant $f_0 = \mathbb{E}F_{\mathbf{f}}$ is determined separately by the one-replica law. $\square$

The restriction to unit diagonal is important: it keeps the marginal Wick ordering fixed while leaving all cross-covariances free. General correlation matrices, rather than a one-parameter equicorrelation curve, are what separate the different loop-free adjacency patterns.

3 Infinite-dimensional Replica–Tannaka reconstruction

We now prove that the loop-free scalar register extracted in [Section 2](#) determines the tensor packet up to one common orthogonal transformation, even when the Gaussian Hilbert space has infinite dimension.

For $q \geq 1$ define the one-leg flattening

$$L_{f_q} : H^{\otimes(q-1)} \longrightarrow H, \quad \langle L_{f_q} u, v \rangle_H = \langle f_q, v \otimes u \rangle_{H^{\otimes q}}. \quad (26)$$

For $q = 1$, the domain is $\mathbb{R}$ and $L_{f_1}(c) = cf_1$. Each L_{f_q} is Hilbert–Schmidt and

$$\|L_{f_q}\|_{\mathcal{S}_2}^2 = \|f_q\|_{H^{\otimes q}}^2. \quad (27)$$

Definition 3.1 (Minimal active support). *The minimal active support of $\mathbf{f}$ is*

$$S(\mathbf{f}) = \overline{\text{span}} \left\{ \text{Ran } L_{f_q} : 1 \leq q \leq m \right\} \subset H. \quad (28)$$

It is the smallest closed subspace $S \subset H$ such that $f_q \in S^{\odot q}$ for all $q \geq 1$.

Indeed, if $v \perp S(\mathbf{f})$, then $L_{f_q}^* v = 0$, so contracting f_q against v in one slot gives zero. Symmetry gives the same in every slot, and hence $f_q \in S(\mathbf{f})^{\odot q}$. The converse is immediate from (26). Directions orthogonal to $S(\mathbf{f})$ are therefore invisible to the polynomial and cannot be reconstructed.

Theorem 3.2 (Scalar Replica–Tannaka reconstruction). *Let H, H' be real separable Hilbert spaces and let $\mathbf{f} = (f_0, \dots, f_m)$ and $\mathbf{f}' = (f'_0, \dots, f'_m)$ be finite symmetric tensor packets on H and H' , respectively. Then*

$$\text{Rep}(\mathbf{f}) = \text{Rep}(\mathbf{f}') \iff \begin{cases} f'_0 = f_0, \\ \text{there is a surjective real unitary } U : S(\mathbf{f}) \rightarrow S(\mathbf{f}') \text{ such that} \\ f'_q = U^{\otimes q} f_q, \quad 1 \leq q \leq m. \end{cases} \quad (29)$$

The unitary need not be unique: the set of all choices is a torsor under the orthogonal stabilizer of the packet.

The proof occupies the rest of this section. We may replace H and H' by their minimal active supports, and therefore assume from now on that $S(\mathbf{f}) = H$ and $S(\mathbf{f}') = H'$. If this support is zero, all nonconstant kernels vanish; the two-replica moments force the same conclusion for the primed packet, and the theorem is immediate. We henceforth treat the nonzero case.

3.1 The support Gram

Define the positive operator

$$D_{\mathbf{f}} = \sum_{q=1}^m L_{f_q} L_{f_q}^* \quad \text{on } H. \quad (30)$$

It is trace class, with

$$\text{Tr } D_{\mathbf{f}} = \sum_{q=1}^m \|f_q\|^2 < \infty. \quad (31)$$

Moreover,

$$\ker D_{\mathbf{f}} = \bigcap_{q=1}^m \ker L_{f_q}^* = S(\mathbf{f})^\perp. \quad (32)$$

Thus $D_{\mathbf{f}}$ is injective in the minimal realization.

Graphically, every summand $L_{f_q} L_{f_q}^*$ is a dipole: two fresh copies of f_q joined along $q - 1$ legs, with one leg left at each end. Consequently, every complete contraction containing powers

of $D_{\mathbf{f}}$ expands into a finite sum of closed loop-free networks made from the original f_q 's. By [Proposition 2.3](#), the scalar replica signature determines all these values.

In particular, equality of replica signatures gives

$$\mathrm{Tr}(D_{\mathbf{f}}^k) = \mathrm{Tr}(D_{\mathbf{f}'}^k), \quad k \geq 1. \quad (33)$$

These moments determine the nonzero spectrum with multiplicity. To see this without invoking a moment theorem with hidden hypotheses, define

$$\mu_{\mathbf{f}} = \sum_{\lambda > 0} m_{\mathbf{f}}(\lambda) \lambda \delta_{\lambda},$$

where $m_{\mathbf{f}}(\lambda)$ is the multiplicity of the nonzero eigenvalue λ of $D_{\mathbf{f}}$. This is a finite measure. Regard both the primed and unprimed measures as measures on the common compact interval $[0, M]$, where $M = \max(\|D_{\mathbf{f}}\|_{\mathrm{op}}, \|D_{\mathbf{f}'}\|_{\mathrm{op}})$; then

$$\int x^j d\mu_{\mathbf{f}}(x) = \mathrm{Tr}(D_{\mathbf{f}}^{j+1}), \quad j \geq 0. \quad (34)$$

Equality of all moments and polynomial density in $C([0, M])$ give $\mu_{\mathbf{f}} = \mu_{\mathbf{f}'}$. The atom at $\lambda > 0$ has mass $m_{\mathbf{f}}(\lambda)\lambda$, so the multiplicities agree. In particular, the two nonzero spectra have the same maximum and

$$\|D_{\mathbf{f}}\|_{\mathrm{op}} = \|D_{\mathbf{f}'}\|_{\mathrm{op}}. \quad (35)$$

3.2 Finite spectral caps recover the missing contractions

We first record the continuity statement which will be used to open self-pairings. It is formulated with enough precision to avoid using an unjustified cut of an infinite-dimensional tensor network.

Lemma 3.3 (Trace-weighted network bound). *Let $G = (V, E)$ be a finite multigraph; loops and parallel edges are allowed. At a vertex v of degree d_v , place a tensor $\tau_v \in H^{\otimes d_v}$. Orient the two ends of every edge e and place a trace-class operator $T_e \in \mathcal{S}_1(H)$ on that edge. The resulting edge-weighted contraction $Z_G((\tau_v), (T_e))$ is absolutely defined and satisfies*

$$|Z_G((\tau_v), (T_e))| \leq \prod_{v \in V} \|\tau_v\|_2 \prod_{e \in E} \|T_e\|_1. \quad (36)$$

It is jointly continuous in the trace norms of the edge operators. More precisely, after fixing an ordering $E = \{e_1, \dots, e_\ell\}$,

$$\begin{aligned} & |Z_G((\tau_v), (T_e)) - Z_G((\tau_v), (S_e))| \\ & \leq \prod_{v \in V} \|\tau_v\|_2 \sum_{j=1}^{\ell} \|T_{e_j} - S_{e_j}\|_1 \prod_{i < j} \|T_{e_i}\|_1 \prod_{i > j} \|S_{e_i}\|_1. \end{aligned} \quad (37)$$

Proof. Choose a nuclear decomposition

$$T_e = \sum_{j \geq 1} s_{e,j} u_{e,j} \otimes v_{e,j}, \quad s_{e,j} \geq 0, \quad \sum_j s_{e,j} = \|T_e\|_1,$$

where the two vector families are orthonormal after zero terms have been discarded. For every choice of one index j_e per edge, insert u_{e,j_e} and v_{e,j_e} into the two endpoint slots. The absolute value of the resulting evaluation at v is at most $\|\tau_v\|_2$, because all inserted endpoint vectors have norm one. The sum of the absolute values of all terms is therefore bounded by

$$\prod_v \|\tau_v\|_2 \sum_{(j_e)_{e \in E}} \prod_{e \in E} s_{e,j_e} = \prod_v \|\tau_v\|_2 \prod_e \|T_e\|_1.$$

This also covers a loop: its two singular vectors are simply inserted in two different slots of the same vertex tensor. Absolute convergence and density of finite-rank operators show that the value is independent of the chosen nuclear decompositions. Finally, replace the edge operators one at a time and apply (36) to each difference. The resulting multilinear telescoping sum is exactly (37). $\square$

Fix $\delta > 0$ outside the common nonzero spectrum and set

$$P_\delta = \mathbf{1}_{[\delta, \infty)}(D_{\mathbf{f}}), \quad P'_\delta = \mathbf{1}_{[\delta, \infty)}(D_{\mathbf{f}'}). \quad (38)$$

These projections have the same finite rank. Although the scalar register directly contains only loop-free graphs, the finite spectral caps allow us to reconstruct contractions with self-pairings.

Lemma 3.4 (Spectral-cap opening). *The common loop-free register determines every complete contraction, with repetitions, of the finite-dimensional packets*

$$\left(P_\delta^{\otimes q} f_q \ (1 \leq q \leq m), \quad B_\delta := P_\delta D_{\mathbf{f}} P_\delta \right) \quad (39)$$

and the analogous primed packet. Corresponding contractions have equal values.

Proof. If $P_\delta = 0$, the assertion is immediate, so assume otherwise. Choose $0 < a < \delta < b$ such that $\sigma(D_{\mathbf{f}}) \cap [a, b] = \emptyset$. There is a continuous function h on the common interval $[0, M]$ introduced above which is zero on $[0, a]$ and equals x^{-1} on $[b, M]$. Functional calculus gives

$$P_\delta = D_{\mathbf{f}} h(D_{\mathbf{f}}). \quad (40)$$

If real polynomials $p_n \rightarrow h$ uniformly and $Q_n = D_{\mathbf{f}} p_n(D_{\mathbf{f}})$, then

$$\|Q_n - P_\delta\|_1 \leq \text{Tr}(D_{\mathbf{f}}) \|p_n - h\|_\infty \rightarrow 0. \quad (41)$$

Take any complete contraction of the tensors in (39). Replace every occurrence of a compressed tensor $P_\delta^{\otimes q} f_q$ by a vertex labelled f_q , and replace every occurrence of $B_\delta = P_\delta D_{\mathbf{f}} P_\delta$ by a two-valent vertex labelled $D_{\mathbf{f}}$. Put one edge weight P_δ on each contracted pair of slots. The projection identity $P_\delta^2 = P_\delta$ shows that the resulting trace-weighted network is exactly the original finite-dimensional contraction, including when an edge pairs two slots of the same vertex.

Replace all of its finitely many edge weights P_δ by Q_n . The trace norms of Q_n are uniformly bounded, and Lemma 3.3, (41), and (37) show that the new network values converge to the desired contraction. This proves continuity without ever applying an unweighted cap to an infinite-dimensional Hilbert tensor.

After expanding the polynomial, every inserted operator is a positive power of $D_{\mathbf{f}}$. Replace each occurrence of $D_{\mathbf{f}}$ by its dipole expansion following (30). A former self-pairing is now a path through fresh tensor vertices; the copies of $D_{\mathbf{f}}$ used as marker vertices are expanded in the same way. No edge joins a vertex to itself. Every approximant is therefore a finite linear combination of closed loop-free networks and is determined by Proposition 2.3. The same approximation is available in the primed system. Equality of the raw registers and the common spectral data imply equality of the limiting contractions. $\square$

3.3 Finite invariant theory and a coherent inverse limit

The finite-dimensional orbit-separation step will be used in the following explicit form.

Proposition 3.5 (Complete contractions separate finite orthogonal tensor orbits). *Let V be a finite-dimensional real inner-product space, let $d_1, \dots, d_s \geq 0$, and put*

$$\mathcal{W} = \bigoplus_{\alpha=1}^s V^{\otimes d_\alpha}.$$

For two points $\tau = (\tau_1, \dots, \tau_s)$ and $\sigma = (\sigma_1, \dots, \sigma_s)$ in $\mathcal{W}$, the following are equivalent:

- (i) $\sigma = U\tau$ for some $U \in O(V)$;
- (ii) *every complete pairwise contraction of every tensor word in the τ_α , with arbitrary repetitions, has the same value for the corresponding word in the σ_α .*

Proof. The implication (i) $\Rightarrow$ (ii) is immediate. For the converse, the first fundamental theorem for the orthogonal group states that the invariant polynomial algebra $\mathbb{R}[\mathcal{W}]^{O(V)}$ is generated by complete pairwise contractions [Wey46, Bra37, Sch08]. Thus (ii) gives equality of every invariant polynomial at τ and σ .

The two $O(V)$ -orbits are compact. If they were distinct, a continuous function on their disjoint union taking the values 0 and 1 on the two components would extend to a continuous function on a sufficiently large closed ball. Polynomial approximation on that ball, followed by averaging over the compact group $O(V)$, would produce an invariant polynomial separating the two orbits. This contradicts equality of all invariant polynomials. Hence the two points lie in one orthogonal orbit. $\square$

Choose any isometry from $P'_\delta H'$ onto $P_\delta H$ and transport the primed compressed packet to the latter space. By Lemma 3.4, all complete contractions of the two finite packets agree. Applying Proposition 3.5 and transporting back gives the isometry below.

It follows from Lemma 3.4 that, for each admissible δ , there exists an isometry

$$U_\delta : P_\delta H \longrightarrow P'_\delta H' \tag{42}$$

such that

$$U_\delta^{\otimes q} P_\delta^{\otimes q} f_q = (P'_\delta)^{\otimes q} f'_q, \quad 1 \leq q \leq m, \tag{43}$$

$$U_\delta B_\delta U_\delta^* = B'_\delta. \tag{44}$$

The marker identity (44) is what makes the finite reconstructions coherent.

Choose strictly decreasing thresholds $\delta_n \downarrow 0$ outside the common nonzero spectrum. Put $H_n = P_{\delta_n} H$ and $H'_n = P'_{\delta_n} H'$. Let $\mathcal{O}_n$ be the compact set of all isometries $V : H_n \rightarrow H'_n$ satisfying (43)–(44) at level n . It is nonempty by the preceding paragraph. If $V \in \mathcal{O}_N$ and $n < N$, then the marker identity and spectral functional calculus imply

$$V(H_n) = H'_n, \quad V|_{H_n} \in \mathcal{O}_n.$$

Indeed, P_{δ_n} is the spectral projection of B_{δ_N} for $[\delta_n, \infty)$, so the marker identity transports it to P'_{δ_n} . Applying these projections in every tensor leg of (43) gives precisely the level- n transport identity.

Consider the compact product $\prod_{n \geq 1} \mathcal{O}_n$. For $n \geq 1$, let C_n be the closed condition

$$V_{n+1}|_{H_n} = V_n.$$

Every finite family $C_1, \dots, C_N$ is satisfiable: choose an element of $\mathcal{O}_{N+1}$, take all its lower restrictions, and choose arbitrary elements at the remaining levels. Compactness and the finite-intersection property give a compatible sequence $(U_n)_{n \geq 1}$.

Because $D_{\mathbf{f}}$ and $D_{\mathbf{f}'}$ are injective, $\bigcup_n H_n$ and $\bigcup_n H'_n$ are dense. The compatible U_n therefore define a surjective unitary $U : H \rightarrow H'$. Finally,

$$P_{\delta_n}^{\otimes q} f_q \longrightarrow f_q, \quad (P'_{\delta_n})^{\otimes q} f'_q \longrightarrow f'_q$$

in Hilbert tensor norm. Passing to the limit in (43) yields

$$U^{\otimes q} f_q = f'_q, \quad 1 \leq q \leq m. \quad (45)$$

This proves the forward implication of [Theorem 3.2](#).

Conversely, suppose (45) holds on the minimal supports and $f'_0 = f_0$. Transport a Σ -correlated isonormal family by U . Equivariance of multiple Wiener integrals gives

$$I_q^{W'}(U^{\otimes q} f_q) = I_q^W(f_q)$$

simultaneously for all replicas. Hence every joint characteristic function in (22) agrees, proving the reverse implication and completing the theorem.

Remark 3.6 (Relation with closed tensor networks). *In finite dimension, converse orthogonal network theorems are closely related to the real Holant theorem; see [You25]. Two additional points are essential here. First, the observer is given only joint laws of one unlabelled scalar Wiener polynomial, and [Proposition 2.3](#) decompresses those laws into the loop-free network register. Second, the support Gram and its finite spectral flags make orbit separation exact in a separable infinite-dimensional Hilbert space.*

3.4 A fixed universal Gaussian canvas

The definition of $\text{Rep}(\mathbf{f})$ permits every correlation matrix. This freedom is convenient for coefficient extraction, but it is far from necessary. We now replace the whole correlation ellipsope by the law of one scalar finite-chaos process driven by a fixed Gaussian covariance kernel. In particular, the covariance intervention is no longer part of the datum.

Put

$$K(x, y) = \exp(-(x - y)^2), \quad x, y \in \mathbb{R}. \quad (46)$$

This is a positive definite kernel with $K(x, x) = 1$ [Sch38]. One direct factorization is obtained in the symmetric Fock space over $\mathbb{R}$:

$$\Psi(x) = e^{-x^2} \bigoplus_{j \geq 0} \frac{2^{j/2}}{\sqrt{j!}} x^{\odot j}, \quad \langle \Psi(x), \Psi(y) \rangle = K(x, y). \quad (47)$$

It is in fact strictly positive definite. If $x_1, \dots, x_r$ are distinct, the first r homogeneous coordinates of the vectors $\Psi(x_i)$ form, up to nonzero row and column factors, the Vandermonde matrix $(x_i^j)_{1 \leq i \leq r, 0 \leq j < r}$. Thus the $\Psi(x_i)$ are linearly independent and their Gram matrix is positive definite.

The kernel is not an arbitrary radial-basis choice. In a Wick network, n parallel edges between two replica vertices replace a correlation $K(x, y)$ by its tensor power $K(x, y)^n$. If one fixed one-dimensional geometry is to internalize this operation, tensor powers should be realized by dilations of the same parameter line. Stationarity, normalization, and smooth nondegeneracy then force both the dilation law and the covariance kernel, up to a change of length scale.

Proposition 3.7 (Rigidity of a replica-stable stationary canvas). *Let $\widetilde{K}$ be a continuous real covariance kernel on $\mathbb{R}$ which is stationary and normalized:*

$$\widetilde{K}(x + a, y + a) = \widetilde{K}(x, y), \quad \widetilde{K}(x, x) = 1.$$

Assume that its stationary profile $\kappa(t) = \widetilde{K}(t, 0)$ is C^2 at zero with $\kappa''(0) < 0$, and that one nontrivial tensor power is represented by some dilation: for an integer $n_0 \geq 2$ and a number $a > 1$,

$$\widetilde{K}(ax, ay) = \widetilde{K}(x, y)^{n_0} \quad (x, y \in \mathbb{R}). \quad (48)$$

Then the dilation and the kernel are both forced:

$$a = \sqrt{n_0}, \quad \widetilde{K}(x, y) = e^{-c(x-y)^2}, \quad c = -\frac{1}{2}\kappa''(0) > 0. \quad (49)$$

Thus rescaling the parameter line reduces the kernel to (46); the curvature normalization $-\kappa''(0) = 2$ fixes it exactly.

Proof. Write $\widetilde{K}(x, y) = \kappa(x - y)$. Symmetry makes κ even. Differentiating (48) twice on the diagonal, and using $\kappa(0) = 1$ and $\kappa'(0) = 0$, gives

$$a^2 \kappa''(0) = n_0 \kappa''(0).$$

The nondegeneracy assumption therefore forces $a = \sqrt{n_0}$.

For every fixed t , the points $a^{-j}t$ eventually lie in a neighborhood on which κ is positive. Iterating (48) backward shows that $\kappa(t) = \kappa(a^{-j}t)^{n_0^j} > 0$, so $\phi = -\log \kappa$ is defined on all of $\mathbb{R}$. Since ϕ is even and C^2 at zero, $\phi(u) = cu^2 + o(u^2)$ with $c = -\kappa''(0)/2 > 0$. Iteration now gives

$$\phi(t) = n_0^j \phi(a^{-j}t) = n_0^j \phi(n_0^{-j/2}t) \longrightarrow ct^2,$$

which proves (49). $\square$

Let $\widetilde{\Psi} : \mathbb{R} \rightarrow \mathcal{G}_{\widetilde{K}}$ be a minimal feature map of $\widetilde{K}$. At the feature-space level, the conclusion of the proposition gives, for every integer $n \geq 1$, that the assignment

$$\widetilde{\Psi}(\sqrt{n}x) \longmapsto \widetilde{\Psi}(x)^{\otimes n} \quad (50)$$

is an isometry between the corresponding cyclic spans: the two families have the same Gram matrices. For (46), $\Psi(x)$ in (47) is the normalized exponential vector at $\sqrt{2}x$. Thus (50) is the exponential law of symmetric Fock space: tensoring replicas becomes dilation of the single canvas coordinate.

Consequently, for every real Hilbert space H there is a jointly Gaussian family $\{\mathbb{W}_x(h) : x \in \mathbb{R}, h \in H\}$ such that

$$\mathbb{E}[\mathbb{W}_x(h)\mathbb{W}_y(k)] = K(x, y) \langle h, k \rangle_H. \quad (51)$$

For instance, take an isonormal process on $\overline{\text{span}}\{\Psi(x) : x \in \mathbb{R}\} \otimes H$ and evaluate it at $\Psi(x) \otimes h$. For every fixed x , the map $h \mapsto \mathbb{W}_x(h)$ is isonormal over H .

For a tensor packet $\mathbf{f}$ as in (18), define the scalar process

$$\mathcal{Z}_{\mathbf{f}}(x) = f_0 + \sum_{q=1}^m I_q^{\mathbb{W}_x}(f_q), \quad x \in \mathbb{R}. \quad (52)$$

The process in (52) has a canonical smooth realization. We record a quantitative form which will also be used in the stability discussion. The space $C(\mathbb{R})$ is always equipped with the topology of locally uniform convergence.

Lemma 3.8 (Smooth realization of the universal canvas). *Let $\mathbf{f}$ and $\mathbf{g}$ be tensor packets of degree at most m on the same real Hilbert space, realized on the same Gaussian canvas. For every $p > 2$ and $R > 0$, their scalar processes admit smooth modifications such that*

$$\mathbb{E} \|\mathcal{Z}_{\mathbf{f}} - \mathcal{Z}_{\mathbf{g}}\|_{C([-R,R])}^p \leq C_{m,p,R} \left(|f_0 - g_0|^2 + \sum_{q=1}^m q! \|f_q - g_q\|^2 \right)^{p/2}. \quad (53)$$

In particular, each process has almost surely C^∞ sample paths. Its law as a $C(\mathbb{R})$ -valued random variable is independent of the chosen continuous modification.

Proof. Realize the full canvas as one isonormal process over the tensor product of H with the feature space in (47). If $h_q = f_q - g_q$, the multiple-integral covariance identity gives

$$\mathbb{E} \left[I_q^{\mathbb{W}_x}(h_q) I_q^{\mathbb{W}_y}(h_q) \right] = q! K(x, y)^q \|h_q\|^2. \quad (54)$$

Distinct chaos orders are orthogonal. Consequently, with $D(x) = \mathcal{Z}_{\mathbf{f}}(x) - \mathcal{Z}_{\mathbf{g}}(x)$,

$$\begin{aligned} \|D(x) - D(y)\|_2^2 &= 2 \sum_{q=1}^m q! \|h_q\|^2 (1 - e^{-q(x-y)^2}) \\ &\leq 2 \sum_{q=1}^m q q! \|h_q\|^2 |x - y|^2. \end{aligned} \quad (55)$$

The increment belongs to the sum of the first m Wiener chaoses. Gaussian hypercontractivity therefore yields, for every $p > 2$,

$$\|D(x) - D(y)\|_p \leq C_{m,p} \left(\sum_{q=1}^m q! \|h_q\|^2 \right)^{1/2} |x - y|, \quad (56)$$

If

$$S^2 = |f_0 - g_0|^2 + \sum_{q=1}^m q! \|h_q\|^2,$$

the same argument gives the correctly scaled pointwise estimate

$$\|D(x)\|_p \leq C_{m,p} S. \quad (57)$$

We next construct one smooth version, rather than unrelated continuous versions for different exponents. Let $\mathcal{G}$ be the feature space generated by the vectors $\Psi(x)$ in (47), let $\mathbb{X}$ be the single isonormal process over $\mathcal{G} \otimes H$ used to realize the canvas, and put $T_x h = \Psi(x) \otimes h$. Then $T_x : H \rightarrow \mathcal{G} \otimes H$ is an isometry, and $T_x^{\otimes q}$ maps $H^{\odot q}$ continuously into $(\mathcal{G} \otimes H)^{\odot q}$. Moreover,

$$I_q^{\mathbb{W}_x}(h_q) = I_q^{\mathbb{X}}(T_x^{\otimes q} h_q). \quad (58)$$

The explicit Fock expansion in (47) shows, by termwise differentiation on compact intervals, that $x \mapsto \Psi(x)$ is C^∞ in the Hilbert norm. Equivalently,

$$\left\| \Psi^{(r)}(x) \right\|^2 = \partial_x^r \partial_y^r K(x, y) \Big|_{y=x} < \infty \quad (r \geq 0). \quad (59)$$

It follows that $F_q(x) := T_x^{\otimes q} h_q$ is a C^∞ curve in $(\mathcal{G} \otimes H)^{\odot q}$, with all derivatives bounded on compact intervals. Since the multiple-integral map is continuous from that Hilbert tensor space to $L^2(\Omega)$, the L^2 derivatives of D are

$$D_r(x) = \mathbf{1}_{\{r=0\}}(f_0 - g_0) + \sum_{q=1}^m I_q^{\mathbb{X}}(F_q^{(r)}(x)), \quad r \geq 0. \quad (60)$$

The mean-value theorem in Hilbert space and hypercontractivity give, on each compact interval, $\|D_r(x) - D_r(y)\|_p \leq C_{m,p,r,R}|x - y|$. Kolmogorov's continuity criterion therefore supplies a continuous modification $\tilde{D}_r$ of every D_r . Choose these modifications simultaneously for the countable family $(r, N) \in \mathbb{N}^2$ on the intervals $[-N, N]$, identifying them on overlaps. Continuity of the multiple-integral map and the fundamental theorem of calculus give in $L^2(\Omega)$

$$D_r(x) - D_r(a) = \int_a^x D_{r+1}(u) du. \quad (61)$$

where the right-hand side is a Bochner integral. Fubini's theorem shows that, for rational a, x , the same identity holds almost surely with D_r, D_{r+1} replaced by $\tilde{D}_r, \tilde{D}_{r+1}$. Intersecting the corresponding countable family of probability-one events and then using continuity extends it to all real a, x . Hence $\tilde{D}_0$ has pathwise derivative $\tilde{D}_1$, and induction proves that it is C^∞ almost surely.

Apply the same construction separately to $\mathcal{Z}_f$ and $\mathcal{Z}_g$. Their difference and the $\tilde{D}_0$ just constructed are continuous modifications of the same process; equality at rational points makes them indistinguishable. We may therefore use these smooth versions in the quantitative estimate below.

Choose s with $1/p < s < 1$. Integrating (56) over $[-R, R]^2$ shows that the expected p th power of the fractional $W^{s,p}$ seminorm of D is bounded by the right-hand side of (53); (57) controls the L^p part. The integral near the diagonal is finite precisely because $s < 1$. The smooth version is jointly measurable; by Fubini it may replace the original process in these fractional Sobolev integrals. The one-dimensional fractional Sobolev embedding $W^{s,p}([-R, R]) \hookrightarrow C^\gamma([-R, R])$ for every $\gamma < s - 1/p$ gives (53) for the smooth version just constructed. Finally, the metric

$$d_C(z, z') = \sum_{j \geq 1} 2^{-j} \min(1, \|z - z'\|_{C([-j, j])}) \quad (62)$$

makes $C(\mathbb{R})$ a Polish space, and its Borel σ -field is generated by evaluation maps at rational points. The smooth version is consequently a measurable $C(\mathbb{R})$ -valued random variable. Two continuous modifications agree almost surely at every rational point and hence everywhere, proving the last assertion. $\square$

From now on, $\mathcal{Z}_f$ denotes this smooth modification.

We next identify the structural reason why restriction to the one-dimensional canvas loses none of the loop-free register. Fix $r \geq 2$ and let

$$E_r = \left\{ x \in \mathbb{R}^r : \sum_{a=1}^r x_a = 0 \right\},$$

the Euclidean configuration space modulo common translations. For a symmetric zero-diagonal array $N = (N_{ab})$ define its weighted graph Laplacian by

$$(L_N)_{aa} = \sum_{b \neq a} N_{ab}, \quad (L_N)_{ab} = -N_{ab} \quad (a \neq b), \quad (63)$$

and put

$$Q_N(x) = x^\top L_N x = \sum_{a < b} N_{ab} (x_a - x_b)^2, \quad \chi_N(x) = e^{-Q_N(x)}. \quad (64)$$

The multiplicative identity $\chi_{N+N'} = \chi_N \chi_{N'}$ makes χ_N the Gaussian character of the multigraph N . The passage $N \mapsto L_N$ is permutation-equivariant and identifies edge weights with translation-invariant quadratic forms on configuration space; it is therefore intrinsic to the replica labels rather than an auxiliary enumeration of graphs.

Lemma 3.9 (Faithfulness of Gaussian graph characters). *For every finite family of distinct nonnegative integral edge arrays N , the functions χ_N are linearly independent on E_r . Equivalently, the homomorphism*

$$\mathbb{R}[T_{ab} : a < b] \longrightarrow C^\omega(E_r), \quad T_{ab} \longmapsto e^{-(x_a - x_b)^2}, \quad (65)$$

is injective.

Proof. The edge forms $(x_a - x_b)^2$, $a < b$, form a basis of $\text{Sym}^2(E_r^*)$. Indeed, their number is $r(r-1)/2 = \dim \text{Sym}^2(E_r^*)$; if a linear combination vanishes on E_r , polarization shows that its symmetric Laplacian A maps E_r into $E_r^\perp$. Since $A\mathbf{1} = 0$, symmetry also gives $A(E_r) \subset E_r$, hence A vanishes on E_r . It vanishes on $\mathbb{R}\mathbf{1}$ as well and is therefore the zero matrix. Its off-diagonal entries then force every coefficient to vanish.

In particular, $N \neq N'$ implies that $Q_N - Q_{N'}$ is a nonzero quadratic form on E_r . Given finitely many arrays, choose $v \in E_r$ outside the zero sets of all these pairwise differences. Then the numbers $\lambda_N = Q_N(v)$ are distinct. A relation $\sum_N c_N \chi_N = 0$, restricted to $x = tv$, becomes

$$\sum_N c_N e^{-\lambda_N t^2} = 0.$$

The exponentials with distinct exponents are linearly independent, as is seen by differentiating in $s = t^2$ at $s = 0$ and using the resulting Vandermonde matrix. Thus every c_N is zero. $\square$

For the one-point statement below, set $E_1 = \{0\}$.

Let $\mathcal{A}_{r,m}$ be the finite set of symmetric zero-diagonal nonnegative integral arrays whose row sums do not exceed m . Equip E_r with its normalized Euclidean Gaussian measure

$$d\gamma_r^\circ(x) = \pi^{-(r-1)/2} e^{-|x|^2} dx. \quad (66)$$

Here dx is the Lebesgue measure induced by the Euclidean structure on E_r . This choice is invariant under permutations of the replica coordinates. Throughout, $\det_{E_r}$ denotes the determinant of the restriction to E_r .

Proposition 3.10 (Canonical Gaussian duality recovers the register). *For a packet $\mathbf{f}$ of degree at most m , define its r -point canvas moment on E_r by*

$$M_{\mathbf{f},r}(x) = \mathbb{E} \prod_{a=1}^r \mathcal{Z}_{\mathbf{f}}(x_a). \quad (67)$$

Then

$$M_{\mathbf{f},r}(x) = \sum_{N \in \mathcal{A}_{r,m}} C_{\mathbf{f}}(N) \chi_N(x), \quad (68)$$

where, writing $q_a = \sum_{b \neq a} N_{ab}$,

$$C_{\mathbf{f}}(N) = \frac{\prod_{a=1}^r q_a!}{\prod_{a < b} N_{ab}!} \text{Contr}_N(f_{q_1}, \dots, f_{q_r}), \quad (69)$$

with an isolated vertex interpreted as the factor f_0 .

More precisely, define the universal matrix and the observed vector

$$G_{N,N'} = \int_{E_r} \chi_N(x) \chi_{N'}(x) d\gamma_r^\circ(x) = \det_{E_r} (I + L_N + L_{N'})^{-1/2}, \quad (70)$$

$$b_{\mathbf{f},r}(N') = \int_{E_r} M_{\mathbf{f},r}(x) \chi_{N'}(x) d\gamma_r^\circ(x). \quad (71)$$

The matrix G is positive definite and

$$(C_{\mathbf{f}}(N))_{N \in \mathcal{A}_{r,m}} = G^{-1}(b_{\mathbf{f},r}(N))_{N \in \mathcal{A}_{r,m}}. \quad (72)$$

Consequently, the $C(\mathbb{R})$ -valued law of the single canvas process determines f_0 and the complete loop-free scalar register.

Proof. For $x \in E_r$, the cross-correlation between vertices a and b is $e^{-(x_a - x_b)^2}$. Wick's formula and [Proposition 2.3](#) therefore give (68) and (69) when all q_a are positive. If $q_a = 0$, vertex a is isolated, contributes the factor f_0 , and may be deleted before applying that proposition. Thus the same formula holds in all cases. The process law determines every joint moment by [Lemma 2.2](#). Moreover, Hölder's inequality gives the uniform bound

$$|M_{\mathbf{f},r}(x)| \leq \|Z_{\mathbf{f}}(0)\|_r^r,$$

so the integrals in (71) are finite.

The Gaussian integral on E_r gives the determinant in (70). By [Lemma 3.9](#), for every nonzero vector (c_N) ,

$$\sum_{N,N'} c_N G_{N,N'} c_{N'} = \int_{E_r} \left| \sum_N c_N \chi_N(x) \right|^2 d\gamma_r^\circ(x) > 0.$$

Thus G is invertible. Taking the inner products of (68) with all the $\chi_{N'}$ gives $b = GC$, which proves (72). The one-point mean separately gives f_0 . $\square$

The decoder is equivariant under every permutation of the replica coordinates. Its faithfulness does not depend on the Gaussian integration measure: any probability measure on E_r with full support would give a positive definite Gram matrix. The Euclidean Gaussian is distinguished here because it is permutation-invariant and yields the closed determinant formula (70). No conditioning estimate for G^{-1} is asserted; the reconstruction is exact rather than numerically stable.

Theorem 3.11 (Smooth universal-canvas Replica–Tannaka theorem). *There is one fixed covariance kernel $K(x, y) = e^{-(x-y)^2}$ on $\mathbb{R}$ and one functorial canvas construction $H \mapsto \mathcal{G}_K \otimes H$, independent of the maximal chaos degree and the tensor packet, with the following property. For a finite symmetric tensor packet $\mathbf{f}$ of degree m on H and a packet $\mathbf{f}'$ of degree m' on H' , put $M = \max(m, m')$ and complete the shorter packet by zero kernels. Then their smooth scalar processes have the same law on $C(\mathbb{R})$ if and only if $f'_0 = f_0$ and there is a surjective real unitary*

$$U : S(\mathbf{f}) \longrightarrow S(\mathbf{f}') \quad \text{such that} \quad f'_q = U^{\otimes q} f_q, \quad 1 \leq q \leq M. \quad (73)$$

Equivalently, the law of one smooth scalar process on one fixed Gaussian canvas is a complete invariant of finite Wiener polynomials modulo their natural orthogonal gauge.

Proof. If the process laws agree, then by [Proposition 3.10](#), the two packets have the same complete loop-free scalar register. The proof of [Theorem 3.2](#) from the support Gram onward uses only this register: it reconstructs all finite spectral compressions, applies finite-dimensional orthogonal orbit separation, and passes to a single unitary by the compact inverse-limit argument. It therefore yields (73).

Conversely, assume (73). For every finite set $x_1, \dots, x_r \in \mathbb{R}$, first restrict the jointly Gaussian isonormal family to the minimal active support, which contains all the kernels, and then transport it by U . Its covariance $K(x_a, x_b) \langle h, k \rangle$ is unchanged, and equivariance of multiple Wiener integrals gives equality of the corresponding scalar vectors. Therefore all finite-dimensional distributions agree. By [Lemma 3.8](#), both processes have continuous versions, and the Borel sigma-field of $C(\mathbb{R})$ with the compact-open topology is generated by evaluations on any countable dense subset. Their $C(\mathbb{R})$ -valued laws agree. $\square$

Proposition 3.12 (Naturality, monoidality, and orbit faithfulness). *The assignment $\mathbf{f} \mapsto \text{Law}_{C(\mathbb{R})}(\mathcal{Z}_{\mathbf{f}})$ has the following properties.*

- (a) *It is orthogonally natural: transporting every tensor by one unitary does not change the law.*
- (b) *It takes orthogonal direct sums of packets to convolution of probability laws on the additive group $C(\mathbb{R})$.*
- (c) *It is continuous, quantitatively on every compact interval, in the Hilbert tensor norms of (53).*
- (d) *On minimal active supports it is orbit-faithful: equal laws imply one common orthogonal transport of every tensor grade.*

Thus it is an orthogonally natural, monoidal, continuous, and orbit-faithful scalarization rather than an abstract measurable encoding. Since the target here is a space of probability laws, this statement does not label distinct transport unitaries; the arrow-level reconstruction for heat packets is the full-faithfulness theorem [Theorem 9.4](#).

Proof. Part (a) is the equivariance of multiple Wiener integrals, and part (c) is [Lemma 3.8](#). For packets $\mathbf{f}$ on H and $\mathbf{g}$ on G , define $\mathbf{f} \boxplus \mathbf{g}$ on $H \oplus G$ by adding the constants and, in every positive grade, adding the tensors embedded from the two orthogonal summands. The canvas isonormal process splits into independent processes on H and G , and

$$\mathcal{Z}_{\mathbf{f} \boxplus \mathbf{g}} = \mathcal{Z}_{\mathbf{f}} + \mathcal{Z}_{\mathbf{g}}.$$

This proves (b). Part (d) is the preceding theorem. □

Remark 3.13 (The scalar compression). *The complete datum is the $C(\mathbb{R})$ -valued law of one smooth scalar process. It replaces the adaptive elliptope of covariance experiments by one stationary, functorial observation architecture: no correlation matrix, tensor label, chaos projection, contraction network, or sampling configuration is supplied to the observer. At replica order r , the unrestricted register has $r(r-1)/2$ off-diagonal coordinates, whereas an r -point canvas configuration modulo translation has only $r-1$ parameters. The Gaussian graph characters remain linearly independent on this lower-dimensional locus, and the determinant Gram decoder recovers every network coefficient exactly. Thus the fixed canvas is a genuine scalar compression of the full orthogonal tensor character.*

Corollary 3.14 (Generic finite configurations are witnesses). *If two finite tensor packets are not equivalent in the sense of [Theorem 3.11](#)—that is, their constants differ or their nonconstant kernels are not related by one common orthogonal transformation on their minimal active supports—then there is an $r \geq 1$ such that, for Lebesgue-almost every $x \in E_r$, the two random vectors*

$$(\mathcal{Z}_{\mathbf{f}}(x_1), \dots, \mathcal{Z}_{\mathbf{f}}(x_r)) \quad \text{and} \quad (\mathcal{Z}_{\mathbf{f}'}(x_1), \dots, \mathcal{Z}_{\mathbf{f}'}(x_r)) \tag{74}$$

have different laws. The separating configurations form an open dense set and include a rational configuration. Thus every inequivalent pair has a finite rational witness on the universal canvas.

Proof. If $f_0 \neq f'_0$, the one-point means differ. Otherwise, after padding the shorter packet by zero kernels, the contrapositive of [Theorem 3.2](#) gives an r for which the two loop-free coefficient vectors differ. By [Lemma 3.9](#),

$$M_{\mathbf{f},r} - M_{\mathbf{f}',r}$$

is then a nonzero real-analytic function on E_r . Its zero set has Lebesgue measure zero and empty interior. Outside that set the mixed moment differs, so the joint laws in (74) differ. The nonzero set is open and dense, hence contains a rational point. □

4 Replicas detect structure invisible to the marginal law

Before turning to geometry, we give an elementary example which shows why the word *correlated* in the replica signature is essential. The example also displays a probabilistic consequence of [Theorem 3.2](#): a one-point law can hide the Hermite rank which controls dependence after a nonlinear Gaussian observation.

Let

$$P(x, y) = x(x^2 + y^2), \quad Q(x, y) = x^3 - 3xy^2. \quad (75)$$

These cubics are not related by an orthogonal change of variables. On the real projective line, the zero set of P consists of the single line $x = 0$, whereas the zero set of $Q = x(x - \sqrt{3}y)(x + \sqrt{3}y)$ consists of three distinct lines. Nevertheless, their ordinary Gaussian laws coincide.

Proposition 4.1 (Same marginal, different replica response). *Let $G = (X, Y)$ be a standard Gaussian vector in $\mathbb{R}^2$. Then*

$$P(G) \stackrel{\text{law}}{=} Q(G). \quad (76)$$

Let $G_1 = (X_1, Y_1)$ and $G_2 = (X_2, Y_2)$ be jointly Gaussian such that the two coordinates are independent, while

$$\mathbb{E}[X_1 X_2] = \mathbb{E}[Y_1 Y_2] = \rho, \quad \mathbb{E}[X_a Y_b] = 0. \quad (77)$$

For $|\rho| < 1$,

$$\mathbb{E}[P(G_1)P(G_2)] = 16\rho + 8\rho^3. \quad (78)$$

$$\mathbb{E}[Q(G_1)Q(G_2)] = 24\rho^3. \quad (79)$$

Thus the two-replica laws differ for every $0 < |\rho| < 1$, even though the one-replica marginal laws are identical.

Proof. Write a standard planar Gaussian in polar form $G = (R \cos \Theta, R \sin \Theta)$, where R and Θ are independent and Θ is uniform modulo 2π . Then

$$P(G) = R^3 \cos \Theta, \quad Q(G) = R^3 \cos(3\Theta).$$

Multiplication by 3 preserves Haar measure on the circle, so $\cos \Theta$ and $\cos(3\Theta)$ have the same distribution and are equally independent of R . This proves [\(76\)](#).

For the replica calculation, let H_j denote the probabilists' Hermite polynomial of degree j . Since $x^3 = H_3(x) + 3H_1(x)$ and $y^2 = H_2(y) + 1$, one has the orthogonal chaos expansions

$$P(x, y) = H_3(x) + H_1(x)H_2(y) + 4H_1(x). \quad (80)$$

$$Q(x, y) = H_3(x) - 3H_1(x)H_2(y). \quad (81)$$

For standard correlated Gaussians Z_1, Z_2 with correlation ρ ,

$$\mathbb{E}[H_j(Z_1)H_k(Z_2)] = \mathbf{1}_{\{j=k\}} j! \rho^j.$$

Coordinate independence and orthogonality of distinct multi-indices therefore give

$$\mathbb{E}[P(G_1)P(G_2)] = 4^2 \rho + (3! + 1!2!) \rho^3 = 16\rho + 8\rho^3,$$

and

$$\mathbb{E}[Q(G_1)Q(G_2)] = (3! + 3^2 1!2!) \rho^3 = 24\rho^3.$$

These identities prove the claim. $\square$

The example has an immediate dependence-theoretic consequence. Recall that the Hermite rank of a centered Gaussian polynomial is the smallest positive chaos order in its Hermite expansion. Equations (80)–(81) show that P has rank 1 and Q has rank 3, although (76) makes that distinction invisible to every statistic based on a single time.

Corollary 4.2 (The same one-time law can hide opposite memory regimes). *Let $(G_n)_{n \in \mathbb{Z}}$ be a stationary centered Gaussian process in $\mathbb{R}^2$ whose coordinates are independent copies of a unit-variance scalar Gaussian process with correlation*

$$r(k) \sim c|k|^{-\alpha}, \quad c > 0, \quad \frac{1}{3} < \alpha < 1.$$

Then the stationary sequences $(P(G_n))$ and $(Q(G_n))$ have the same one-time marginal distribution, but

$$\sum_{k \in \mathbb{Z}} |\text{Cov}(P(G_0), P(G_k))| = \infty, \quad \sum_{k \in \mathbb{Z}} |\text{Cov}(Q(G_0), Q(G_k))| < \infty. \quad (82)$$

Proof. For all sufficiently large $|k|$, one has $|r(k)| < 1$, so apply (78) and (79) with $\rho = r(k)$. The finitely many remaining lags do not affect summability. The first covariance is asymptotic to $16c|k|^{-\alpha}$ and is not absolutely summable because $\alpha < 1$. The second is asymptotic to $24c^3|k|^{-3\alpha}$ and is absolutely summable because $3\alpha > 1$. $\square$

Thus equal marginals can conceal not merely a tensorial coordinate choice but a qualitative change in temporal dependence. The replica parameter probes the covariance intervention which exposes the missing Hermite grade. In the manifold and gauge applications below, the same principle exposes multiplication, support bands, and ultimately local geometry which the quadratic spectrum alone cannot see.

5 A scalar Gaussian observable hears the manifold

We now show that a single scalar polynomial of a heat-regularized white noise determines the full Riemannian isometry class, provided that one is allowed to observe all its scalar correlated replicas. The quadratic chaos recovers the heat operator, whereas one further even chaos recovers the order structure of the underlying function space. Their combination turns the abstract orthogonal operator supplied by Replica–Tannaka rigidity into pullback by a Riemannian isometry.

Throughout this section, a closed Riemannian manifold means a compact smooth manifold without boundary. If (M, g) is such a manifold, we write dv_g for its Riemannian measure and use the nonnegative convention

$$\Delta_g = -\text{div}_g \nabla_g$$

for the Laplace–Beltrami operator on $H_M := L^2(M, dv_g; \mathbb{R})$. Its heat kernel is denoted by $p_s^M(x, y)$, so that

$$(e^{-s\Delta_g} h)(x) = \int_M p_s^M(x, y) h(y) dv_g(y), \quad s > 0.$$

5.1 The observable and its replica signature

Fix $t > 0$ and an even integer $q \geq 4$, and set

$$A_{M,t} := e^{-t\Delta_g/2}, \quad k_{M,t,x} := p_{t/2}^M(x, \cdot) \in H_M. \quad (83)$$

Let W_M be an isonormal Gaussian process over the real Hilbert space H_M ; thus

$$\mathbb{E}[W_M(h)W_M(\ell)] = \langle h, \ell \rangle_{H_M}.$$

The heat-regularized white noise is the smooth Gaussian field

$$X_{M,t}(x) := W_M(k_{M,t,x}). \quad (84)$$

It has covariance

$$\mathbb{E}[X_{M,t}(x)X_{M,t}(y)] = \langle k_{M,t,x}, k_{M,t,y} \rangle_{H_M} = p_t^M(x, y). \quad (85)$$

The asserted smoothness holds in the samplewise sense. Indeed, for every integer $s \geq 0$ the spectral theorem gives

$$\mathbb{E} \|X_{M,t}\|_{H^s(M)}^2 = \text{Tr}((1 + \Delta_g)^s e^{-t\Delta_g}) < \infty.$$

Taking the countable intersection over s and using Sobolev embedding shows that $X_{M,t} \in C^\infty(M)$ almost surely.

We use the multiple-Wiener-integral normalization

$$\mathbb{E}[I_r(f)I_s(h)] = \mathbf{1}_{\{r=s\}} r! \langle f, h \rangle_{H_M^{\odot r}}. \quad (86)$$

For $\sigma_{M,t}(x)^2 := p_t^M(x, x)$ and the probabilists' Hermite polynomial He_r , the pointwise Wick power is

$$: X_{M,t}(x)^r := \sigma_{M,t}(x)^r \text{He}_r\left(\frac{X_{M,t}(x)}{\sigma_{M,t}(x)}\right) = I_r(k_{M,t,x}^{\otimes r}). \quad (87)$$

In particular,

$$\begin{aligned} : X_{M,t}(x)^2 : &= X_{M,t}(x)^2 - \sigma_{M,t}(x)^2, \\ : X_{M,t}(x)^6 : &= X_{M,t}(x)^6 - 15\sigma_{M,t}(x)^2 X_{M,t}(x)^4 \\ &\quad + 45\sigma_{M,t}(x)^4 X_{M,t}(x)^2 - 15\sigma_{M,t}(x)^6. \end{aligned}$$

Define the scalar observable

$$F_{M,t,q} := \int_M (: X_{M,t}(x)^2 : + : X_{M,t}(x)^q :) dv_g(x). \quad (88)$$

There is no renormalization hidden in this definition beyond (87). Indeed, the map $x \mapsto k_{M,t,x}$ is continuous from M to H_M , because

$$\|k_{M,t,x} - k_{M,t,y}\|_{H_M}^2 = p_t^M(x, x) + p_t^M(y, y) - 2p_t^M(x, y).$$

It follows from compactness that, for every $r \geq 1$, the Bochner integral

$$f_{M,t}^{(r)} := \int_M k_{M,t,x}^{\otimes r} dv_g(x) \in H_M^{\odot r} \quad (89)$$

is well defined. More explicitly,

$$\int_M \|k_{M,t,x}^{\otimes r}\| dv_g(x) = \int_M p_t^M(x, x)^{r/2} dv_g(x) < \infty.$$

By (86), the map $f \mapsto I_r(f)$ is continuous from $H_M^{\odot r}$ to $L^2(\Omega)$. Consequently the integral in (88) is a Bochner integral in $L^2(\Omega)$ and

$$F_{M,t,q} = I_2(f_{M,t}^{(2)}) + I_q(f_{M,t}^{(q)}). \quad (90)$$

For $r \geq 1$, let

$$\text{Corr}_r^\circ := \{\Sigma \in \mathbb{R}^{r \times r} : \Sigma = \Sigma^\top > 0, \Sigma_{aa} = 1\}.$$

Given $\Sigma \in \text{Corr}_r^\circ$, let $(W_M^{(1)}, \dots, W_M^{(r)})$ be a replicated isonormal family satisfying

$$\mathbb{E}[W_M^{(a)}(h)W_M^{(b)}(\ell)] = \Sigma_{ab}\langle h, \ell \rangle_{H_M}. \quad (91)$$

Let $F_{M,t,q}^{(a)}$ be (88) constructed from $W_M^{(a)}$. The *scalar replica signature* of (M, g) at (t, q) is the indexed family

$$\Phi_{M,t,q}(r, \Sigma, \theta) := \mathbb{E} \exp \left(i \sum_{a=1}^r \theta_a F_{M,t,q}^{(a)} \right), \quad \mathfrak{R}_{t,q}(M, g) := (\Phi_{M,t,q}(r, \Sigma, \theta))_{r \geq 1, \Sigma \in \text{Corr}_r^\circ, \theta \in \mathbb{R}^r}, \quad (92)$$

where the indexing by (r, Σ, θ) is part of the datum.

By Theorem 3.2, equality of these signatures yields one and the same real unitary on the minimal active support which transports both homogeneous kernels. The common-unitary conclusion is essential: it would not follow from the two marginal chaos laws taken separately.

Theorem 5.1 (A scalar Gaussian observable hears the manifold). *Let (M, g) and (N, h) be closed connected Riemannian manifolds, let $t > 0$, and let $q \geq 4$ be even. Then*

$$\mathfrak{R}_{t,q}(M, g) = \mathfrak{R}_{t,q}(N, h) \iff (M, g) \simeq (N, h) \text{ as Riemannian manifolds.} \quad (93)$$

Thus the minimal even choice within this norm-rigidity mechanism is $q = 4$: the correlated scalar replicas of

$$\int_M (: X_{M,t}(x)^2 : + : X_{M,t}(x)^4 :) dv_g(x)$$

determine the unpointed Riemannian isometry class of (M, g) . The same statement holds for every even $q > 2$, including $q = 6$.

Proof. We divide the nontrivial implication into five explicit steps.

Step 1: one common Replica–Tannaka unitary.

Assume $\mathfrak{R}_{t,q}(M, g) = \mathfrak{R}_{t,q}(N, h)$. Apply Theorem 3.2 to the chaos decompositions (90). It gives a real unitary U between their minimal one-particle supports such that

$$f_{N,t}^{(2)} = U^{\odot 2} f_{M,t}^{(2)}, \quad f_{N,t}^{(q)} = U^{\odot q} f_{M,t}^{(q)}. \quad (94)$$

These supports are in fact all of H_M and H_N . To verify this, define the Hilbert–Schmidt operator $C_M : H_M \rightarrow H_M$ by

$$\langle C_M a, b \rangle_{H_M} := \langle f_{M,t}^{(2)}, b \otimes a \rangle_{H_M^{\otimes 2}}. \quad (95)$$

For $a, b \in H_M$, Fubini’s theorem and (83) give

$$\begin{aligned} \langle C_M a, b \rangle_{H_M} &= \int_M \langle k_{M,t,x}, a \rangle \langle k_{M,t,x}, b \rangle dv_g(x) \\ &= \langle A_{M,t} a, A_{M,t} b \rangle_{H_M} = \langle e^{-t\Delta_g} a, b \rangle_{H_M}. \end{aligned} \quad (96)$$

Hence $C_M = e^{-t\Delta_g}$. This operator is injective and has dense range; the one-particle support of $f_{M,t}^{(2)}$, which is the closure of the range of its flattening, is therefore H_M . The same argument applies to N . Thus

$$U : H_M \longrightarrow H_N \text{ is a surjective real unitary.} \quad (97)$$

Step 2: the quadratic chaos recovers the full heat semigroup. Flattening the first identity in (94) and using (96) yields

$$U e^{-t\Delta_g} U^{-1} = e^{-t\Delta_h}. \quad (98)$$

There is no loss in passing from one heat time to the generator. Indeed, $e^{-t\Delta_g}$ and $e^{-t\Delta_h}$ are positive injective contractions, and the Borel functional calculus gives

$$\Delta_g = -\frac{1}{t} \log(e^{-t\Delta_g}), \quad \Delta_h = -\frac{1}{t} \log(e^{-t\Delta_h}). \quad (99)$$

Here the value of $\log$ at 0 is immaterial because the spectral projection of either heat operator at $\{0\}$ is zero; the domains in (99) are the corresponding functional-calculus domains. Therefore

$$U \operatorname{Dom}(\Delta_g) = \operatorname{Dom}(\Delta_h), \quad U\Delta_g = \Delta_h U, \quad (100)$$

and, for every $s \geq 0$,

$$Ue^{-s\Delta_g} = e^{-s\Delta_h}U. \quad (101)$$

In particular, $UA_{M,t} = A_{N,t}U$.

Step 3: the even q th chaos turns U into an L^q -isometry. For every $a \in H_M$, the definition (89) gives the exact diagonal identity

$$\langle f_{M,t}^{(q)}, a^{\otimes q} \rangle = \int_M (A_{M,t}a)(x)^q dv_g(x) = \|A_{M,t}a\|_{L^q(M)}^q, \quad (102)$$

where the last equality uses that q is even. Evaluate the second identity in (94) against $(Ua)^{\otimes q}$ and use $UA_{M,t} = A_{N,t}U$. We obtain

$$\|UA_{M,t}a\|_{L^q(N)} = \|A_{M,t}a\|_{L^q(M)} \quad (a \in H_M). \quad (103)$$

The range of $A_{M,t}$ is dense in $L^q(M)$. One direct verification is that it contains every finite linear combination of Laplace eigenfunctions, while such combinations are dense in $C^\infty(M)$ in its Fréchet topology and hence dense in $L^q(M)$. Similarly, $\operatorname{Ran}(A_{N,t})$ is dense in $L^q(N)$, and (101) gives

$$U \operatorname{Ran}(A_{M,t}) = \operatorname{Ran}(A_{N,t}).$$

It follows from (103) that U extends uniquely to a surjective real linear isometry

$$\tilde{U} : L^q(M, dv_g) \longrightarrow L^q(N, dv_h). \quad (104)$$

Since both measures are finite and $q > 2$, convergence in L^q implies convergence in L^2 . Thus the extension $\tilde{U}$ agrees with the original L^2 -unitary U on all of $L^q(M)$; below we use the same symbol U for both realizations.

Step 4: Lamperti rigidity makes U an order isomorphism, up to one global sign. The Banach–Lamperti theorem [Lam58] for the surjective L^q -isometry (104), with $q \neq 2$, supplies a measure-class isomorphism $\phi : N \rightarrow M$ modulo null sets and a nonvanishing measurable real function a on N such that

$$(Uf)(y) = a(y)f(\phi(y)) \quad \text{for almost every } y \in N, \quad f \in L^q(M). \quad (105)$$

Applying the L^q -isometry and the L^2 -unitarity to indicator functions gives, for every measurable $E \subseteq M$,

$$\int_{\phi^{-1}(E)} |a|^q dv_h = dv_g(E), \quad \int_{\phi^{-1}(E)} |a|^2 dv_h = dv_g(E). \quad (106)$$

Because ϕ is an isomorphism of measure algebras, (106) implies $|a|^q = |a|^2$ almost everywhere. Since a is nonzero and $q > 2$, we have $|a| = 1$ almost everywhere, and then

$$dv_h(\phi^{-1}(E)) = dv_g(E). \quad (107)$$

Moreover, $a = U\mathbf{1}_M$. By (100), $a \in \ker \Delta_h$. Elements of this kernel are smooth and, because N is connected, constant. Therefore $a = c$ almost everywhere for one $c \in \{-1, 1\}$. Replacing U by $V := cU$ gives

$$(Vf)(y) = f(\phi(y)), \quad V\mathbf{1}_M = \mathbf{1}_N. \quad (108)$$

The operator V is a unitary order isomorphism and still intertwines the Laplacians. Since both tensor degrees 2 and q are even, this harmless global sign change also preserves both identities in (94).

Step 5: the order isomorphism is pullback by a Riemannian isometry. On a closed manifold, elliptic regularity and the spectral theorem give

$$C^\infty(M) = \bigcap_{m \geq 1} \text{Dom}(\Delta_g^m), \quad C^\infty(N) = \bigcap_{m \geq 1} \text{Dom}(\Delta_h^m). \quad (109)$$

It follows from (100) that V restricts to a bijection $C^\infty(M) \rightarrow C^\infty(N)$. The representation (108) shows that this restriction is a unital algebra isomorphism.

For completeness, we now remove the “almost everywhere” qualifier from ϕ . Choose a smooth embedding $\iota = (\iota_1, \dots, \iota_L) : M \hookrightarrow \mathbb{R}^L$ and define

$$\Phi : N \longrightarrow \mathbb{R}^L, \quad \Phi(y) := (V\iota_1(y), \dots, V\iota_L(y)).$$

The map Φ is smooth and equals $\iota \circ \phi$ almost everywhere. Since $\iota(M)$ is closed, continuity of Φ and the fact that every nonempty open subset of N has positive Riemannian volume imply $\Phi(N) \subseteq \iota(M)$. Hence

$$\bar{\phi} := \iota^{-1} \circ \Phi : N \longrightarrow M$$

is smooth and agrees with ϕ almost everywhere. Applying the same construction to V^{-1} yields a smooth map $\bar{\psi} : M \rightarrow N$. The two compositions agree almost everywhere with the identity maps; by continuity they agree everywhere. Thus $\bar{\phi}$ is a diffeomorphism, with inverse $\bar{\psi}$, and

$$Vf = f \circ \bar{\phi} \quad (f \in C^\infty(M)). \quad (110)$$

It remains only to recover the metric. For the nonnegative Laplacian set

$$\Gamma_g(f_1, f_2) := \frac{1}{2}(f_1\Delta_g f_2 + f_2\Delta_g f_1 - \Delta_g(f_1 f_2)) = \langle \nabla f_1, \nabla f_2 \rangle_g. \quad (111)$$

Because V is both multiplicative and Laplacian-intertwining,

$$\Gamma_h(Vf_1, Vf_2) = V\Gamma_g(f_1, f_2) \quad (f_1, f_2 \in C^\infty(M)). \quad (112)$$

At $y \in N$, with $x = \bar{\phi}(y)$, this says

$$h_y^{-1}((d\bar{\phi}_y)^* df_1(x), (d\bar{\phi}_y)^* df_2(x)) = g_x^{-1}(df_1(x), df_2(x)). \quad (113)$$

The differentials of smooth functions span T_x^*M , so (113) says that $(d\bar{\phi}_y)^* : T_x^*M \rightarrow T_y^*N$ is an isometry for the cometrics. Equivalently, $d\bar{\phi}_y$ is an isometry for the tangent metrics. Thus $\bar{\phi} : (N, h) \rightarrow (M, g)$ is a Riemannian isometry.

We finally prove the converse. Let $\bar{\phi} : (N, h) \rightarrow (M, g)$ be a Riemannian isometry and define $V : H_M \rightarrow H_N$ by $Vf = f \circ \bar{\phi}$. Then V is a real unitary, $V\Delta_g = \Delta_h V$, and $VA_{M,t} = A_{N,t}V$. For $r \in \{2, q\}$ and $a_1, \dots, a_r \in H_M$, we have

$$\left\langle f_{N,t}^{(r)}, Va_1 \otimes \dots \otimes Va_r \right\rangle$$

$$\begin{aligned}
&= \int_N \prod_{j=1}^r (A_{N,t} V a_j)(y) dv_h(y) \\
&= \int_M \prod_{j=1}^r (A_{M,t} a_j)(x) dv_g(x) = \langle f_{M,t}^{(r)}, a_1 \otimes \cdots \otimes a_r \rangle.
\end{aligned}$$

Hence

$$f_{N,t}^{(r)} = V^{\odot r} f_{M,t}^{(r)}, \quad r \in \{2, q\}. \quad (114)$$

For every replicated family over H_M , define a replicated family over H_N by $W_N^{(a)}(b) := W_M^{(a)}(V^{-1}b)$. It has exactly the covariance (91). Unitary equivariance of multiple Wiener integrals and (114) give

$$F_{N,t,q}^{(a)} = F_{M,t,q}^{(a)} \quad \text{almost surely for every replica } a.$$

All the indexed joint characteristic functions (92) are therefore equal. This proves the reverse implication and completes the proof. $\square$

The freely indexed family of correlation matrices in (92) can be replaced by the one fixed Gaussian canvas constructed in Section 3.4. Indeed, let $(\mathbb{W}_{M,x})_{x \in \mathbb{R}}$ be that canvas over H_M and put

$$\mathcal{Z}_{M,t,q}(x) := I_2^{\mathbb{W}_{M,x}}(f_{M,t}^{(2)}) + I_q^{\mathbb{W}_{M,x}}(f_{M,t}^{(q)}), \quad x \in \mathbb{R}. \quad (115)$$

Corollary 5.2 (One fixed Gaussian canvas hears the manifold). *Let (M, g) and (N, h) be closed connected Riemannian manifolds, let $t > 0$, and let $q \geq 4$ be even. Then*

$$\text{Law}_{C(\mathbb{R})}(\mathcal{Z}_{M,t,q}) = \text{Law}_{C(\mathbb{R})}(\mathcal{Z}_{N,t,q}) \iff (M, g) \simeq (N, h). \quad (116)$$

Here both smooth processes use the same covariance kernel $K(x, y) = e^{-(x-y)^2}$ on the whole real line; it depends on neither the manifolds, nor t , nor q .

Proof. Equality of the two process laws and Theorem 3.11 give a single surjective real unitary $U : H_M \rightarrow H_N$ satisfying

$$f_{N,t}^{(2)} = U^{\odot 2} f_{M,t}^{(2)}, \quad f_{N,t}^{(q)} = U^{\odot q} f_{M,t}^{(q)}.$$

The active supports are the full L^2 spaces by (96). Steps 2–5 in the proof of Theorem 5.1 therefore produce a Riemannian isometry. Conversely, a Riemannian isometry transports the two kernels by (114); the converse implication of Theorem 3.11 then identifies the canvas laws. $\square$

Remark 5.3 (Why both chaos orders are essential). *The quadratic kernel alone recovers $e^{-t\Delta_g}$, hence the unitary equivalence class of Δ_g , but this is only spectral information and does not distinguish isospectral nonisometric manifolds. The even q th kernel forces the same intertwining unitary to preserve an L^q norm. Since $q \neq 2$, Banach–Lamperti rigidity recovers the measure algebra and its order; the Laplacian then recovers the smooth structure and the metric through (112).*

Remark 5.4 (Coefficients and the choice of heat convention). *The theorem remains true for*

$$\alpha \int_M : X_{M,t}(x)^2 : dv_g(x) + \beta \int_M : X_{M,t}(x)^q : dv_g(x), \quad \alpha\beta \neq 0,$$

provided that the known coefficients α and β are kept fixed across manifolds. The convention $X_{M,t} = W_M(p_{t/2}^M(x, \cdot))$ was chosen so that its covariance is p_t^M and the quadratic tensor flattens exactly to $e^{-t\Delta_g}$, with no factor 2 in (96).

6 The same scalar process hears singular diffusions

The preceding argument is not tied to smooth coordinates. Its analytic core only uses a symmetric heat semigroup, an L^4 multiplication tensor, and the fact that an order isomorphism intertwining diffusion recovers the intrinsic geometry. We formulate that core at the level of Dirichlet spaces and then specialize it to compact RCD spaces.

6.1 An abstract Dirichlet-space theorem

For $i \in \{1, 2\}$, let (X_i, μ_i) be a finite standard Borel measure space and let $(P_s^{(i)})_{s \geq 0}$ be a strongly continuous symmetric Markov semigroup on $L^2(X_i, \mu_i)$, with nonnegative self-adjoint generator L_i . Fix a marked time $t > 0$ and assume:

(D1) $P_s^{(i)}$ is conservative and irreducible;

(D2) $P_{t/2}^{(i)}$ has a symmetric measurable kernel $p_{t/2}^{(i)}(x, y)$ whose row map $x \mapsto p_{t/2}^{(i)}(x, \cdot) \in L^2(X_i, \mu_i)$ is strongly measurable and which represents $P_{t/2}^{(i)}$;

(D3)

$$\int_{X_i} \left(\int_{X_i} |p_{t/2}^{(i)}(x, y)|^2 d\mu_i(y) \right)^2 d\mu_i(x) < \infty; \quad (117)$$

(D4) the bounded map $P_{t/2}^{(i)} : L^2(X_i, \mu_i) \rightarrow L^4(X_i, \mu_i)$ has dense range in $L^4(X_i, \mu_i)$.

Put $H_i = L^2(X_i, \mu_i)$ and

$$k_{i,t,x} = p_{t/2}^{(i)}(x, \cdot) \in H_i. \quad (118)$$

Put $\kappa_{i,t}(x) = \|k_{i,t,x}\|_2^2$. Assumption (117) says precisely that $\kappa_{i,t} \in L^2(X_i, \mu_i)$. Together with finiteness of μ_i , this makes the following Bochner integrals well defined:

$$Q_{i,t} = \int_{X_i} k_{i,t,x}^{\otimes 2} d\mu_i(x) \in H_i^{\odot 2}, \quad C_{i,t} = \int_{X_i} k_{i,t,x}^{\otimes 4} d\mu_i(x) \in H_i^{\odot 4}. \quad (119)$$

Indeed,

$$\int_{X_i} \|k_{i,t,x}\|_2^2 d\mu_i(x) \leq \mu_i(X_i)^{1/2} \left(\int_{X_i} \kappa_{i,t}(x)^2 d\mu_i(x) \right)^{1/2}, \quad (120)$$

$$\int_{X_i} \|k_{i,t,x}\|_2^4 d\mu_i(x) = \int_{X_i} \kappa_{i,t}(x)^2 d\mu_i(x). \quad (121)$$

Let $\mathbb{W}_{i,u}$, $u \in \mathbb{R}$, be the universal Gaussian canvas over H_i from [Section 3.4](#); thus

$$\mathbb{E}[\mathbb{W}_{i,u}(f)\mathbb{W}_{i,v}(g)] = e^{-(u-v)^2} \langle f, g \rangle_{H_i}. \quad (122)$$

Define the scalar process

$$\mathcal{Y}_{i,t}(u) = I_2^{\mathbb{W}_{i,u}}(Q_{i,t}) + I_4^{\mathbb{W}_{i,u}}(C_{i,t}), \quad u \in \mathbb{R}. \quad (123)$$

Theorem 6.1 (A scalar canvas process reconstructs a diffusion). *Under assumptions (D1)–(D4), the following are equivalent.*

- (i) *The two smooth processes in (123) have the same law on $C(\mathbb{R})$.*

(ii) There is a measure-preserving measure-algebra isomorphism $\phi : X_2 \rightarrow X_1$ such that the unitary

$$Vf = f \circ \phi \quad (124)$$

satisfies

$$VP_s^{(1)} = P_s^{(2)}V, \quad s \geq 0. \quad (125)$$

(iii) The pullback V in (124) identifies the Dirichlet forms:

$$V \text{Dom}(\mathcal{E}_1) = \text{Dom}(\mathcal{E}_2), \quad \mathcal{E}_2(Vf, Vg) = \mathcal{E}_1(f, g). \quad (126)$$

Here $\mathcal{E}_i(f, g) = \langle L_i^{1/2}f, L_i^{1/2}g \rangle$.

Proof. Assume first that the process laws agree. Apply [Theorem 3.11](#) to the graded packet $(0, 0, Q_{i,t}, 0, C_{i,t})$. We obtain a common real unitary $U : H_1 \rightarrow H_2$ satisfying

$$U^{\otimes 2}Q_{1,t} = Q_{2,t}, \quad U^{\otimes 4}C_{1,t} = C_{2,t}. \quad (127)$$

The unitary acts on the whole L^2 space. Indeed, the flattening of the quadratic tensor is

$$\begin{aligned} \langle \text{Flat}(Q_{i,t})f, g \rangle_{H_i} &= \int_{X_i} (P_{t/2}^{(i)}f)(x)(P_{t/2}^{(i)}g)(x) d\mu_i(x) \\ &= \langle P_t^{(i)}f, g \rangle_{H_i}, \end{aligned} \quad (128)$$

so $\text{Flat}(Q_{i,t}) = P_t^{(i)}$. The operator $P_t^{(i)} = e^{-tL_i}$ is injective and has dense range.

Flattening the first equality in (127) gives

$$UP_t^{(1)} = P_t^{(2)}U. \quad (129)$$

The unbounded Borel functional calculus applied to $L_i = -t^{-1} \log P_t^{(i)}$ yields $UL_1 = L_2U$ on the generator domains, and consequently

$$UP_s^{(1)} = P_s^{(2)}U, \quad s \geq 0. \quad (130)$$

The quartic tensor identifies the L^4 norm. For every $f \in H_i$,

$$\langle C_{i,t}, f^{\otimes 4} \rangle = \int_{X_i} (P_{t/2}^{(i)}f)(x)^4 d\mu_i(x) = \|P_{t/2}^{(i)}f\|_{L^4(X_i)}^4. \quad (131)$$

Using (127) and (130), we obtain

$$\|UP_{t/2}^{(1)}f\|_{L^4(X_2)} = \|P_{t/2}^{(1)}f\|_{L^4(X_1)}. \quad (132)$$

Assumption (D4) on both spaces therefore extends U to a surjective L^4 -isometry. Since the measures are finite, $L^4 \hookrightarrow L^2$, and the extension agrees with the original L^2 -unitary on L^4 .

The real Banach–Lamperti theorem now gives a measure-class isomorphism $\phi : X_2 \rightarrow X_1$ and a nonvanishing measurable function a such that

$$Uf = a(f \circ \phi). \quad (133)$$

Apply both the L^2 and L^4 norm identities to indicator functions. Exactly as in (106), the resulting measure identities imply $|a|^2 = |a|^4$, hence $|a| = 1$, and show that ϕ preserves measure. Moreover, $a = U\mathbf{1}_{X_1}$ is fixed by every $P_s^{(2)}$ by conservativity and (130). Irreducibility implies that the fixed space consists of constants. Thus $a \equiv \varepsilon$ for one $\varepsilon \in \{-1, 1\}$. The unitary $V = \varepsilon U$ is precisely the pullback in (124), and it still intertwines all heat times. This proves (ii).

The equivalence between (125) and (126) follows from the spectral theorem: intertwining the generators is equivalent to intertwining their square roots and domains. Thus (ii) and (iii) are equivalent.

Conversely, suppose (ii). The identities $VP_{t/2}^{(1)} = P_{t/2}^{(2)}V$ and preservation of measure give $V^{\otimes 2}Q_{1,t} = Q_{2,t}$ and $V^{\otimes 4}C_{1,t} = C_{2,t}$. The reverse implication in [Theorem 3.11](#) now gives equality of the two process laws, proving (i). $\square$

6.2 Compact RCD spaces

Proposition 6.2 (RCD heat flows satisfy the reconstruction hypotheses). *Let (X, d, μ) be a compact connected $\text{RCD}(K, N)$ space with full support and $N < \infty$. For every marked time $t > 0$, its Cheeger heat semigroup satisfies (D1)–(D4) of [Theorem 6.1](#).*

Proof. The space is a compact standard Borel space with finite measure. Its Cheeger energy is a regular strongly local quadratic Dirichlet form, and the associated heat semigroup $(P_s)_{s \geq 0}$ is symmetric, conservative, and Markovian [[AGS14a](#), [AGS14b](#), [EKS15](#), [Gig18](#), [FOT11](#)]. Connectedness and strict positivity of the heat kernel give irreducibility.

For $s > 0$, the heat kernel $p_s(x, y)$ is symmetric and jointly measurable. The Gaussian estimates on finite-dimensional RCD spaces imply that $p_s(x, x)$ is bounded on the compact space [[JLZ16](#)]. The semigroup identity gives

$$\left\| p_{t/2}(x, \cdot) \right\|_2^2 = p_t(x, x). \quad (134)$$

Consequently,

$$\int_X \left\| p_{t/2}(x, \cdot) \right\|_2^4 d\mu(x) = \int_X p_t(x, x)^2 d\mu(x) < \infty, \quad (135)$$

which is (D3). Moreover, Cauchy–Schwarz and (134) yield

$$\left\| P_{t/2}f \right\|_\infty \leq \sup_{x \in X} p_t(x, x)^{1/2} \|f\|_2, \quad (136)$$

so $P_{t/2} : L^2 \rightarrow L^4$ is bounded.

It remains to prove density of its L^4 range. Let $g \in L^{4/3}(X, \mu)$ annihilate $P_{t/2}(L^2)$. By symmetry and the boundedness just proved, the Banach adjoint is the heat operator $P_{t/2} : L^{4/3} \rightarrow L^2$, and the annihilation identity gives $P_{t/2}g = 0$. The heat semigroup is analytic on $L^{4/3}$ [[BGL14](#)]. Hence $P_sg = 0$ for every $s > t/2$ by the semigroup law, analyticity propagates this identity to all $s > 0$, and strong continuity at the origin gives $g = 0$. The annihilator of $P_{t/2}(L^2)$ in $L^{4/3}$ is therefore trivial, proving (D4). The remaining hypotheses are the symmetry, conservativity, irreducibility, and measurable kernel properties already verified. $\square$

The carré du champ of the Cheeger form is the squared minimal weak upper gradient. The Sobolev-to-Lipschitz property identifies the intrinsic Dirichlet metric with d . Therefore an order isomorphism intertwining the Cheeger heat semigroups preserves the intrinsic metric and is represented by a measure-preserving isometry; equivalently, this is the RCD specialization of the rigidity theorem of Lenz, Schmidt and Wirth [[LSW18](#)].

Corollary 6.3 (One scalar process reconstructs every compact RCD space). *Fix $t > 0$. Let (X_i, d_i, μ_i) , $i \in \{1, 2\}$, be compact connected $\text{RCD}(K_i, N_i)$ spaces with full support and $N_i < \infty$. Construct the universal canvas process (123) from the Cheeger heat flow at time t . Then*

$$\text{Law}_{C(\mathbb{R})}(\mathcal{Y}_{1,t}) = \text{Law}_{C(\mathbb{R})}(\mathcal{Y}_{2,t}) \quad (137)$$

if and only if there is a measure-preserving isometry

$$\phi : (X_2, d_2, \mu_2) \longrightarrow (X_1, d_1, \mu_1). \quad (138)$$

No common synthetic dimension bound, common curvature bound, or regular-set hypothesis is required for this exact reconstruction.

Proof. By [Proposition 6.2](#), equality of the two scalar laws gives a measure-preserving order isomorphism V intertwining the Cheeger heat semigroups and identifying their Dirichlet forms. The order structure transports the carré du champ, while the Sobolev-to-Lipschitz property identifies the induced intrinsic metrics with d_1 and d_2 . The resulting representative of the measure algebra map is therefore a measure-preserving isometry. Conversely, a measure-preserving isometry identifies the Cheeger energies, heat semigroups, heat kernels, and hence the two tensors in [\(119\)](#); the universal-canvas theorem then identifies the process laws. $\square$

Remark 6.4 (A larger nonsmooth reconstruction class). *The abstract diffusion theorem applies to every finite irreducible symmetric Dirichlet geometry satisfying (D2)–(D4). For quasi-regular forms the reconstructed order isomorphism is realized by a quasi-homeomorphism, and for regular strongly local forms it preserves the intrinsic metric whenever that metric generates the given topology [LSW18]. Compact finite-dimensional RCD spaces provide the canonical metric-measure realization of this general scalar reconstruction principle.*

7 Scalar reconstruction of orthogonal gauge geometry

The preceding application used a scalar heat equation. We now show that the reconstruction mechanism is not confined to scalar geometry. For a real Euclidean vector bundle equipped with a metric connection, the flexible correlated-replica signature of one scalar quadratic–quartic Wick observable reconstructs the base manifold, the bundle, and the connection up to the unavoidable joint action of isometries and orthogonal gauge transformations. By [Theorem 3.11](#), the same information is equivalently encoded by the law of one smooth scalar process on our fixed Gaussian canvas; this is the final form of the result.

7.1 Heat-smoothed Gaussian sections

Let $(E, \langle \cdot, \cdot \rangle_E) \rightarrow (M, g)$ be a smooth real Euclidean vector bundle of positive finite rank over a closed connected manifold, and let ∇^E be a metric connection. Put

$$H_E = L^2(M, E; dv_g), \quad L_E = (\nabla^E)^* \nabla^E, \quad A_{E,t} = e^{-tL_E/2}. \quad (139)$$

The operator L_E is nonnegative, self-adjoint, and elliptic, whereas $A_{E,t}$ is injective, smoothing, and has dense range.

Let W_E be an isonormal process over H_E . The random section $X_{E,t} = A_{E,t}W_E$ is understood by duality, or pointwise through the smooth kernel of $A_{E,t}$, and has a C^∞ modification. Let

$$K_{t,x} : H_E \longrightarrow E_x, \quad K_{t,x}s = (A_{E,t}s)(x),$$

and choose an orthonormal basis $(\varepsilon_1(x), \dots, \varepsilon_r(x))$ of E_x . Define

$$k_{x,\alpha} = K_{t,x}^* \varepsilon_\alpha(x), \quad q_{t,x} = \sum_{\alpha=1}^r k_{x,\alpha}^{\otimes 2} \in H_E^{\odot 2}. \quad (140)$$

Equivalently,

$$q_{t,x} = (K_{t,x}^* \otimes K_{t,x}^*)(\langle \cdot, \cdot \rangle_E^{-1}(x)).$$

The tensor $q_{t,x}$ is independent of the chosen orthonormal basis: an orthogonal change of basis cancels after summing the two matrix indices. Its symmetric square is likewise intrinsic. Set

$$Q_{E,t} = \int_M q_{t,x} dv_g(x), \quad C_{E,t} = \int_M (q_{t,x} \odot q_{t,x}) dv_g(x). \quad (141)$$

These Bochner integrals belong respectively to $H_E^{\odot 2}$ and $H_E^{\odot 4}$: smoothness of the heat kernel gives continuity of the integrands in Hilbert tensor norm, and M is compact.

The invariant homogeneous Wick polynomials are defined tensorially by

$$: |X_{E,t}(x)|^2 : = I_2(q_{t,x}), \quad : |X_{E,t}(x)|^4 : = I_4(q_{t,x} \odot q_{t,x}). \quad (142)$$

In a local orthonormal frame, the second expression is the full joint Wick ordering of $\sum_{\alpha,\beta} X_\alpha^2 X_\beta^2$. This definition is therefore frame-independent even when the pointwise covariance is not a scalar endomorphism. The scalar gauge observable is

$$\mathcal{G}_{E,\nabla,t} := \int_M (: |X_{E,t}(x)|^2 : + : |X_{E,t}(x)|^4 :) dv_g(x) = I_2(Q_{E,t}) + I_4(C_{E,t}). \quad (143)$$

Its complete correlated-replica signature, denoted $\mathfrak{R}_t(M, g, E, \nabla^E)$, is defined as in (92), by coupling the underlying isonormal processes with every finite unit-diagonal positive definite correlation matrix.

Two identities explain the choice. For $s_1, s_2, s \in H_E$,

$$\langle Q_{E,t}, s_1 \otimes s_2 \rangle = \int_M \langle A_{E,t}s_1, A_{E,t}s_2 \rangle_E dv_g = \langle e^{-tL_E} s_1, s_2 \rangle_{H_E}, \quad (144)$$

$$\langle C_{E,t}, s^{\otimes 4} \rangle = \int_M |A_{E,t}s(x)|_E^4 dv_g(x) = \|A_{E,t}s\|_{L^4(M,E)}^4. \quad (145)$$

Thus the quadratic flattening is the connection heat operator, while the quartic tensor contains the non-Hilbertian L^4 geometry.

7.2 A vector-valued Lamperti lemma

We need the spatial form of a map which is simultaneously an L^2 -unitary and an L^4 -isometry. This is a bundle version of the vector-valued Banach–Lamperti theorem [Sou78, GJ87]; in our finite-rank Euclidean setting it has the following direct proof.

Lemma 7.1 (Simultaneous L^2 – L^4 rigidity). *Let $E \rightarrow M$ and $F \rightarrow N$ be positive-rank smooth Euclidean vector bundles over closed manifolds. Suppose a real linear map $U_2 : L^2(M, E) \rightarrow L^2(N, F)$ is unitary and a real linear map $U_4 : L^4(M, E) \rightarrow L^4(N, F)$ is a surjective isometry. Suppose there is a linear subspace $\mathcal{C} \subset L^2(M, E) \cap L^4(M, E)$ which is dense for the norm $\|\cdot\|_2 + \|\cdot\|_4$, whose image has the same property on N , and on which $U_2 = U_4$. Write U for their common restriction to $L^2 \cap L^4$. Then there exist a measure-preserving measure-algebra isomorphism $\phi : N \rightarrow M$ modulo null sets and a measurable family of surjective fiber isometries $J_y : E_{\phi(y)} \rightarrow F_y$ such that*

$$(Us)(y) = J_y s(\phi(y)) \quad \text{for a.e. } y \in N. \quad (146)$$

In particular, the two bundles have the same rank.

Proof. First, U_2 and U_4 agree on all of $L^2 \cap L^4$. Indeed, approximate any section in the sum norm by elements of $\mathcal{C}$. The two image sequences converge to $U_2 s$ in L^2 and to $U_4 s$ in L^4 ; after passing to a common almost-everywhere convergent subsequence, the limits coincide. The same argument applies to the inverses.

For vectors a, b in a real Hilbert space, direct expansion gives

$$|a + b|^4 + |a - b|^4 - 2|a|^4 - 2|b|^4 = 4|a|^2|b|^2 + 8\langle a, b \rangle^2. \quad (147)$$

After integration, the left-hand side vanishes exactly when two sections have disjoint essential supports. Hence U and U^{-1} preserve disjointness.

For measurable $A \subset M$, put

$$\mathcal{B}_A = \{s \in L^4(M, E) : s = 0 \text{ a.e. on } M \setminus A\}.$$

These are precisely the bands determined by disjointness: if $\mathcal{S}^\perp$ denotes all sections disjoint from every member of $\mathcal{S}$, then $\mathcal{S}^{\perp\perp} = \mathcal{B}_A$, where A is the essential union of the supports of members of $\mathcal{S}$. Therefore U induces a Boolean isomorphism of the measure algebras. Since compact manifolds are standard Borel finite measure spaces, it is represented by a measure-class isomorphism $\phi : N \rightarrow M$, uniquely modulo null sets.

The decomposition into the complementary bands of A and $M \setminus A$ gives

$$U(\mathbf{1}_A s) = \mathbf{1}_{\phi^{-1}(A)} U s. \quad (148)$$

By linearity and uniform approximation by simple functions,

$$U(bs) = (b \circ \phi) U s, \quad b \in L^\infty(M). \quad (149)$$

Choose a countable Borel trivialization of E by orthonormal frames. On one trivializing set A , let the frame be $e_1, \dots, e_r$ and put $u_i = U(\mathbf{1}_A e_i)$. Formula (149) gives, first for simple sections and then by density,

$$U\left(\sum_{i=1}^r b_i e_i\right)(y) = \sum_{i=1}^r b_i(\phi(y)) u_i(y), \quad y \in \phi^{-1}(A). \quad (150)$$

Thus U is represented locally by a measurable linear map $T_y : E_{\phi(y)} \rightarrow F_y$.

Let $\rho > 0$ be characterized by

$$\int_{\phi^{-1}(B)} \rho(y) dv_h(y) = dv_g(B) \quad (B \subset M \text{ measurable}). \quad (151)$$

Apply the L^4 -isometry to $\mathbf{1}_B c$, with c constant in the chosen frame. Varying B and then using a countable dense set of vectors gives

$$|T_y c|^4 = \rho(y) |c|^4 \quad (c \in E_{\phi(y)}) \quad (152)$$

almost everywhere. The L^2 -unitarity applied to the same sections gives

$$|T_y c|^2 = \rho(y) |c|^2. \quad (153)$$

Squaring the latter and comparing with the former yields $\rho^2 = \rho$. Since $\rho > 0$, we have $\rho = 1$. Hence ϕ preserves volume and T_y is a fiberwise isometric embedding. The same argument for U^{-1} shows that it is onto and that the ranks agree. Taking $J_y = T_y$ proves the lemma. $\square$

7.3 Reconstruction of the connection

Theorem 7.2 (A scalar replica observable hears an orthogonal gauge field). *Fix $t > 0$. Let (M, g, E, ∇^E) and (N, h, F, ∇^F) be smooth positive-rank Euclidean vector bundles with metric connections over closed connected Riemannian manifolds. No equality of dimensions or ranks is assumed. Then*

$$\mathfrak{R}_t(M, g, E, \nabla^E) = \mathfrak{R}_t(N, h, F, \nabla^F) \quad (154)$$

if and only if there exist a Riemannian isometry $\phi : (N, h) \rightarrow (M, g)$ and a smooth fiberwise orthogonal bundle isomorphism $J : \phi^ E \rightarrow F$ which transports the connection:*

$$\nabla^F(J(\phi^* s)) = (\text{Id}_{T^*N} \otimes J)(\phi^*(\nabla^E s)) \quad (s \in C^\infty(M, E)). \quad (155)$$

Thus the signature determines the bundle topology, metric connection, curvature, and holonomy up to the natural isometry- $O(r)$ gauge action.

Proof. Assume (154). Apply Theorem 3.2 to the tensor pairs $(Q_{E,t}, C_{E,t})$ and $(Q_{F,t}, C_{F,t})$. By (144), the quadratic flattenings are injective with dense range, so both minimal active supports are the whole L^2 spaces. We obtain a surjective real unitary $U : H_E \rightarrow H_F$ such that

$$Ue^{-tL_E} = e^{-tL_F}U, \quad U^{\odot 4}C_{E,t} = C_{F,t}. \quad (156)$$

The unbounded logarithm in the spectral functional calculus gives

$$U \operatorname{Dom}(L_E) = \operatorname{Dom}(L_F), \quad UL_E = L_FU, \quad Ue^{-sL_E} = e^{-sL_F}U \quad (s \geq 0). \quad (157)$$

Evaluate the quartic identity in (156) on four copies of $s \in H_E$ and use (145). Since $Ue^{-tL_E/2} = e^{-tL_F/2}U$,

$$\|UA_{E,t}s\|_{L^4(N,F)} = \|A_{E,t}s\|_{L^4(M,E)}. \quad (158)$$

Let $(\psi_j)_{j \geq 1}$ be an orthonormal eigensection basis of L_E , with $L_E\psi_j = \lambda_j\psi_j$. For every smooth section, its spectral truncations converge in every Sobolev norm by elliptic regularity, hence in L^4 . Since smooth sections are dense in L^4 , finite eigensection sums are dense there. Every such sum belongs to the heat range because

$$\sum_{j=1}^m c_j \psi_j = A_{E,t} \left(\sum_{j=1}^m e^{t\lambda_j/2} c_j \psi_j \right).$$

Consequently $\operatorname{Ran}(A_{E,t})$ is L^4 -dense, and heat intertwining gives $U \operatorname{Ran}(A_{E,t}) = \operatorname{Ran}(A_{F,t})$. Equation (158) therefore extends uniquely to a surjective L^4 -isometry. Finite spectral sums are also dense in $L^2 \cap L^4$ for the sum norm: first approximate simultaneously by a smooth section, then use convergence of its spectral truncations in a high Sobolev norm. Thus the L^2 and L^4 realizations satisfy the common-core hypothesis of Lemma 7.1. Consequently,

$$(Us)(y) = J_y s(\phi(y)) \quad (159)$$

for a volume-preserving measure-class isomorphism ϕ and a measurable field of fiber isometries J_y .

We next prove smoothness. Elliptic regularity gives

$$C^\infty(M, E) = \bigcap_{k \geq 1} \operatorname{Dom}(L_E^k), \quad C^\infty(N, F) = \bigcap_{k \geq 1} \operatorname{Dom}(L_F^k). \quad (160)$$

Thus U bijects the smooth section spaces. For $b \in C^\infty(M)$, (149) gives

$$UM_bU^{-1} = M_{b \circ \phi}. \quad (161)$$

The left-hand side preserves smooth sections. Around any point of N , choose a smooth section v of F which is nowhere zero. Then

$$b \circ \phi = \frac{\langle M_{b \circ \phi} v, v \rangle_F}{|v|_F^2}$$

is smooth there. Applying the argument to U^{-1} shows that ϕ and its inverse pull smooth functions to smooth functions. To make the resulting regularization explicit, choose a Whitney embedding $\iota : M \hookrightarrow \mathbb{R}^K$. Every coordinate of $\iota \circ \phi$ has a smooth representative on N , and these representatives define a smooth map $\Phi : N \rightarrow \mathbb{R}^K$. It lands in $\iota(M)$ almost everywhere. Since $\iota(M)$ is closed and every nonempty open subset of N has positive volume, continuity implies $\Phi(N) \subset \iota(M)$. Thus ϕ has a smooth representative $\iota^{-1} \circ \Phi$. Repeating the construction for ϕ^{-1} gives a smooth map in the other direction. Their two compositions equal the identity almost everywhere and hence everywhere by continuity. The representatives are therefore

inverse diffeomorphisms; this argument in particular recovers the dimension. If $e_1, \dots, e_r$ is a local smooth orthonormal frame of E and χ is a cutoff equal to one on a smaller patch, the smooth sections $U(\chi e_i)$ are locally the columns of J . Hence J has a smooth fiberwise orthogonal representative.

The principal symbol of the Bochner Laplacian is

$$\sigma_2(L_E)(x, \xi) = |\xi|_g^2 \text{Id}_{E_x}. \quad (162)$$

Conjugating by the zeroth-order bundle map J over ϕ and using $UL_E = L_F U$ yields $|(d\phi_y)^* \xi|_{h_y^*}^2 = |\xi|_{g_{\phi(y)}^*}^2$ for every $y \in N$ and $\xi \in T_{\phi(y)}^* M$. Thus ϕ is a Riemannian isometry.

Transport ∇^E by (ϕ, J) to a metric connection $\tilde{\nabla}$ on F . Naturality and (157) give

$$\tilde{\nabla}^* \tilde{\nabla} = (\nabla^F)^* \nabla^F. \quad (163)$$

Write $\nabla^F = \tilde{\nabla} + B$, with $B \in \Omega^1(N, \mathfrak{so}(F))$. At a point y , choose normal coordinates and a $\tilde{\nabla}$ -normal orthonormal bundle frame. The first-order part at y of the difference in (163) is

$$-2 \sum_{j=1}^{\dim N} B(X_j) \partial_{X_j}, \quad (164)$$

where $X_1, \dots, X_{\dim N}$ is the chosen orthonormal tangent frame; all remaining terms have order zero. Equality of the differential operators forces $B(X_j) = 0$ for every j . Thus $B = 0$, which is exactly (155).

Conversely, suppose (ϕ, J) satisfies (155). Then $(Us)(y) = J_y s(\phi(y))$ defines a real L^2 -unitary which intertwines the connection heat operators and preserves fiber metric and volume. Formula (141) gives $U^{\odot 2} Q_{E,t} = Q_{F,t}$ and $U^{\odot 4} C_{E,t} = C_{F,t}$. Equivariance of multiple Wiener integrals identifies every correlated joint law. $\square$

Corollary 7.3 (A smooth matrix potential is heard as well). *Let $V_E \in C^\infty(M, \text{Sym}(E))$ and $V_F \in C^\infty(N, \text{Sym}(F))$, and replace the Bochner Laplacians by*

$$\mathcal{L}_E = (\nabla^E)^* \nabla^E + V_E, \quad \mathcal{L}_F = (\nabla^F)^* \nabla^F + V_F.$$

Equality of the corresponding replica signatures is equivalent to the existence of (ϕ, J) as in Theorem 7.2, together with

$$V_F(y) = J_y V_E(\phi(y)) J_y^{-1}. \quad (165)$$

Proof. Each $\mathcal{L}$ is self-adjoint, bounded below, elliptic, and has a smoothing heat semigroup. Replica–Tannaka and the preceding L^4 argument first produce a smooth spatial unitary U satisfying $U \mathcal{L}_E U^{-1} = \mathcal{L}_F$. Comparison of principal symbols shows that the underlying diffeomorphism is a Riemannian isometry. Transport ∇^E to a metric connection $\tilde{\nabla}$ on F and write $\nabla^F = \tilde{\nabla} + B$. The potentials are zeroth-order operators, so the first-order part of the difference is

$$-2 \sum_j B(X_j) \partial_{X_j}.$$

It follows that $B = 0$. The remaining zeroth-order equality is precisely (165); the converse is immediate. $\square$

The same reduction to one predetermined Gaussian canvas is available for the vector-bundle observable. Let $(\mathbb{W}_{E,x})_{x \in \mathbb{R}}$ be the canvas of Section 3.4 over H_E and define

$$\mathcal{Z}_{E,\nabla,V,t}(x) := I_2^{\mathbb{W}_{E,x}}(Q_{E,t}) + I_4^{\mathbb{W}_{E,x}}(C_{E,t}), \quad x \in \mathbb{R}. \quad (166)$$

For $V = 0$ these are the tensors in (141); for a potential, they are formed in exactly the same way with $A_{E,t} = e^{-t((\nabla^E)^* \nabla^E + V_E)/2}$.

Corollary 7.4 (One fixed Gaussian canvas hears the gauge field). *Let (M, g, E, ∇^E, V_E) and (N, h, F, ∇^F, V_F) be as in Corollary 7.3; the case $V_E = V_F = 0$ is included. Then*

$$\text{Law}_{C(\mathbb{R})}(\mathcal{Z}_{E, \nabla^E, V_E, t}) = \text{Law}_{C(\mathbb{R})}(\mathcal{Z}_{F, \nabla^F, V_F, t}) \quad (167)$$

if and only if there exist a Riemannian isometry $\phi : (N, h) \rightarrow (M, g)$ and a smooth orthogonal bundle isomorphism $J : \phi^ E \rightarrow F$ such that*

$$\nabla^F = J\phi^*\nabla^E J^{-1}, \quad V_F = J(\phi^* V_E)J^{-1}. \quad (168)$$

The covariance kernel $K(x, y) = e^{-(x-y)^2}$ on $\mathbb{R}$ is fixed independently of all geometric data.

Proof. By Theorem 3.11, equality in (167) is equivalent to the existence of a single surjective real unitary $U : H_E \rightarrow H_F$ transporting both $Q_{E,t}$ and $C_{E,t}$. The quadratic flattening is the injective operator $e^{-t\mathcal{L}^E}$, so the active support is all of H_E , and similarly on F . The proof of Theorem 7.2 and Corollary 7.3, starting from this common unitary, yields precisely (168). Conversely, such (ϕ, J) transports the two tensors, and the converse part of the universal-canvas theorem gives equality of the process laws. $\square$

7.4 Gauge moduli, holonomy, and finite separation

The reconstruction theorem acts on the full moduli space of smooth metric connections and therefore on every geometrically defined locus inside it.

Corollary 7.5 (Gauge moduli, holonomy, and finite scalar separation). *Fix $t > 0$.*

- (a) *On a fixed closed connected (M, g) and at every rank $r \geq 1$, the universal-canvas law induces an injection from smooth metric connections on all rank- r Euclidean bundles, modulo base isometries and orthogonal gauge. Its restriction to the Yang–Mills locus $d_{\nabla}^* F_{\nabla} = 0$ is injective as well.*
- (b) *The law determines the curvature tensor, the Yang–Mills energy, and the complete parallel-transport functor up to the same equivalence. For every piecewise smooth path γ in N , equivalent data satisfy*

$$P_{\nabla^F}(\gamma) = J_{\gamma(1)} P_{\nabla^E}(\phi \circ \gamma) J_{\gamma(0)}^{-1}. \quad (169)$$

In particular, it determines the conjugacy class of holonomy along every loop.

- (c) *On the flat locus, equality of canvas laws is equivalent to isometric conjugacy of the monodromy representations. For a fixed base, this gives an injection*

$$\frac{\text{Hom}(\pi_1(M), O(r))/O(r)}{\text{Isom}(M, g)} \hookrightarrow \{\text{universal scalar canvas laws at time } t\}. \quad (170)$$

For $r = 1$, the invariant is the class in $H^1(M; \mathbb{Z}_2)$ modulo the natural isometry action.

- (d) *Let $\mathcal{L}_E = (\nabla^E)^* \nabla^E + V_E$ and $\mathcal{L}_F = (\nabla^F)^* \nabla^F + V_F$ be two smooth self-adjoint matrix Schrödinger geometries which are not isometry–orthogonal-gauge equivalent. Then a finite rational canvas configuration separates their joint scalar laws. This finite separation persists for every ordinary isospectral but geometrically inequivalent pair.*

Proof. The gauge reconstruction theorem with potential identifies equality of canvas laws with a unique isometry–orthogonal-gauge arrow. Curvature naturality gives

$$F_{\nabla^F} = (\text{Id}_{\Lambda^2 T^* N} \otimes \text{Ad } J) \phi^* F_{\nabla^E},$$

which preserves the Yang–Mills equation and curvature energy, while naturality of covariant differentiation along paths gives (169). For flat connections, parallel orthogonal bundle isomorphism is equivalent to conjugacy of monodromy, proving (170). The finite rational separation follows from Corollary 3.14 applied to the inequivalent master packets. Ordinary isospectrality fixes only the quadratic spectral orbit and leaves the quartic spatial tensor available to separate the pair. $\square$

Remark 7.6 (Why orthogonal gauge is optimal). *The radial observable does not mark an orientation or a complex structure on the fibers. This is a genuine obstruction, not a defect of the proof. Choose a closed connected Riemannian manifold (M, g) of dimension at least two with trivial isometry group, and a one-form $A \in \Omega^1(M)$ satisfying $dA \neq 0$. Such metrics exist, for example among generic metrics on a closed surface of genus at least two. On the trivial oriented rank-two bundle put*

$$J_0 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad \nabla^A = d + J_0 A.$$

A constant reflection $C \in O(2) \setminus SO(2)$ satisfies $CJ_0C^{-1} = -J_0$, hence conjugates ∇^A to ∇^{-A} and preserves every scalar replica signature. They are not equivalent by a base isometry and an $SO(2)$ gauge. Indeed, the base isometry must be the identity, and an $SO(2)$ gauge changes A by a closed one-form with integral periods. Sending A to $-A$ would therefore force $-2A$ to be closed, contrary to $dA \neq 0$. Therefore a reconstruction modulo $SO(r)$, $U(r)$, or a smaller structure group requires additional Tannakian marker tensors defining that reduction. The theorem modulo $O(r)$ is sharp.

Remark 7.7 (Relation with inverse connection problems). *Connection Laplacians have been reconstructed from local hyperbolic Dirichlet-to-Neumann data [KOP18]. On a fixed closed negatively curved manifold with simple length spectrum, ordinary Bochner spectra determine low-rank bundles and connections up to gauge [CL25]. The datum here is different: the observer sees only the law of one smooth scalar process on the fixed Gaussian canvas of Section 3.4. The canvas compresses the flexible correlated-replica signature into a single fixed-architecture law with no adaptive covariance intervention, while the quartic chaos supplies the spatial L^4 module structure missing from the ordinary spectrum.*

8 Deterministic completeness of the master heat packet

The preceding reconstruction theorems admit a formulation which no longer mentions random variables. The complete invariant is a family of ordinary scalar contractions of two heat tensors. This also makes the residual symmetry of the invariant completely explicit.

8.1 The two heat tensors and their closed networks

Let

$$\mathfrak{X} = (M, g, E, \nabla^E, V_E), \quad \mathcal{L}_{\mathfrak{X}} = (\nabla^E)^* \nabla^E + V_E, \quad (171)$$

where M is closed and connected, $E \rightarrow M$ is a positive-rank real Euclidean bundle, ∇^E is a metric connection, and $V_E \in C^\infty(M, \text{Sym}(E))$. Fix $t > 0$ and put

$$H_{\mathfrak{X}} = L^2(M, E), \quad A_{\mathfrak{X}, t} = e^{-t\mathcal{L}_{\mathfrak{X}}/2}, \quad R_{\mathfrak{X}, t, x} s = (A_{\mathfrak{X}, t} s)(x). \quad (172)$$

The evaluation map in (172) is bounded because the heat operator is smoothing. If $(\varepsilon_\alpha)_{\alpha=1}^{\text{rank } E}$ is any orthonormal basis of E_x , define

$$q_{\mathfrak{X}, t, x} := \sum_{\alpha} (R_{\mathfrak{X}, t, x}^* \varepsilon_\alpha)^{\otimes 2} \in H_{\mathfrak{X}}^{\odot 2}. \quad (173)$$

As in (140), this does not depend on the chosen basis. The two integrated radial heat tensors are

$$T_{\mathfrak{X},t}^{(2)} := \int_M q_{\mathfrak{X},t,x} dv_g(x), \quad T_{\mathfrak{X},t}^{(4)} := \int_M q_{\mathfrak{X},t,x} \odot q_{\mathfrak{X},t,x} dv_g(x). \quad (174)$$

Thus these are $Q_{E,t}$ and $C_{E,t}$ from (141), with the same definition when a potential is present. In particular, the flattening of $T_{\mathfrak{X},t}^{(2)}$ is the injective operator $e^{-t\mathcal{L}_{\mathfrak{X}}}$.

Definition 8.1 (The master heat packet). *The canonical marked-time packet of $\mathfrak{X}$ is*

$$\mathbf{T}_t^{\text{master}}(\mathfrak{X}) := (H_{\mathfrak{X}}, T_{\mathfrak{X},t}^{(2)}, T_{\mathfrak{X},t}^{(4)}). \quad (175)$$

It is called master because it is defined directly from heat evaluation and every homogeneous scalar realization constructed below factors through it.

Proposition 8.2 (Minimality of the positive radial master packet). *Let V be a nonzero finite-dimensional real Euclidean space. For every integer $d \geq 0$, the space of homogeneous $O(V)$ -invariant polynomials of degree d is*

$$\mathbb{R}[V]_d^{O(V)} = \begin{cases} \{0\}, & d \text{ odd}, \\ \mathbb{R}|v|^{2k}, & d = 2k. \end{cases} \quad (176)$$

Consequently, among local homogeneous scalar observables of one vector that are intrinsic under arbitrary orthogonal changes of fiber frame, degree two gives only the Hilbert norm and degree four is the first additional positive degree producing a non-Hilbertian L^p geometry. The heat packet $\mathbf{T}_t^{\text{master}} = (T_t^{(2)}, T_t^{(4)})$ is therefore minimal among positive radial packets that work uniformly for every finite fiber rank.

Proof. The orthogonal group acts transitively on every Euclidean sphere, so an invariant homogeneous polynomial P of degree d is constant on the unit sphere. Homogeneity gives $P(v) = c|v|^d$ for $v \neq 0$. If d is odd, invariance under $-\text{Id}$ forces $P = 0$. If $d = 2k$, then $P(v) = c|v|^{2k}$, which is polynomial and spans the invariant space. For $d = 2$, the integrated heat observable is exactly the quadratic tensor whose flattening is the Hilbert-space heat operator. The next positive radial degree is $d = 4$, and its diagonal is the fourth power of the L^4 norm on the heat range. $\square$

Remark 8.3 (The native cubic scalar sector). *For the trivial real line bundle one may also form*

$$T_{M,t}^{(3)} = \int_M k_{M,t,x}^{\otimes 3} dv_g(x).$$

If one orthogonal map $U : L^2(M) \rightarrow L^2(N)$ transports $T_{M,t}^{(2)}$ and $T_{M,t}^{(3)}$, the quadratic identity intertwines the heat semigroups and the cubic identity gives

$$\int_N (Uf)(Ug)(Uh) dv_h = \int_M fgh dv_g$$

on the heat range. Since $U\mathbf{1} = c\mathbf{1}$, the L^2 and cubic identities evaluated on constants give $c = 1$. Unitarity then yields

$$\langle U(fg), Uh \rangle_{L^2(N)} = \langle (Uf)(Ug), Uh \rangle_{L^2(N)},$$

so $U(fg) = (Uf)(Ug)$ on the smooth spectral core and hence on $C^\infty(M)$. Thus the scalar sector has a native cubic multiplication realization. The master status of the quadratic–quartic packet is stronger: it is positive, radial, and intrinsic for bundles of every rank and for the singular Dirichlet setting.

A *closed loop-free 2/4-network* is a finite multigraph $\Gamma = (\mathbf{V}, \mathbf{E})$ without self-loops, with parallel edges allowed, such that every vertex has degree 2 or 4. At a vertex v of degree $d(v)$ place $T_{\mathfrak{x},t}^{(d(v))}$ and contract tensor slots along the edges. Symmetry of the tensors makes the resulting number independent of all choices of slot order. We denote it by

$$\mathcal{N}_{\Gamma,t}(\mathfrak{X}) := \text{Contr}_{\Gamma} \left((T_{\mathfrak{x},t}^{(d(v))})_{v \in \mathbf{V}} \right). \quad (177)$$

Equivalently, if $\tau_{2,x} = q_{\mathfrak{x},t,x}$ and $\tau_{4,x} = q_{\mathfrak{x},t,x} \odot q_{\mathfrak{x},t,x}$, then Bochner–Fubini gives the local heat-kernel formula

$$\mathcal{N}_{\Gamma,t}(\mathfrak{X}) = \int_{M^{\mathbf{V}}} \text{Contr}_{\Gamma} \left((\tau_{d(v),x_v})_{v \in \mathbf{V}} \right) \prod_{v \in \mathbf{V}} dv_g(x_v). \quad (178)$$

For the trivial real line bundle with its trivial connection and $V_E = 0$, this reduces to the familiar scalar partition function

$$\mathcal{N}_{\Gamma,t}(M, g) = \int_{M^{\mathbf{V}}} \prod_{\{v,w\} \in \mathbf{E}} p_t^M(x_v, x_w) \prod_{v \in \mathbf{V}} dv_g(x_v), \quad (179)$$

where edges are counted with multiplicity. In the vector-bundle case, each edge carries the kernel of $e^{-t\mathcal{L}_{\mathfrak{X}}}$ and the radial tensors perform the intrinsic fiber-index contractions at the vertices.

8.2 The master equivalence

Theorem 8.4 (Master heat-packet equivalence). *Fix $t > 0$. Let $\mathfrak{X} = (M, g, E, \nabla^E, V_E)$ and $\mathfrak{Y} = (N, h, F, \nabla^F, V_F)$ be data of the form (171); no equality of base dimensions or bundle ranks is assumed. The following are equivalent.*

- (i) *There exist a Riemannian isometry $\phi : (N, h) \rightarrow (M, g)$ and a smooth orthogonal bundle isomorphism $J : \phi^*E \rightarrow F$ such that*

$$\nabla^F = J\phi^*\nabla^E J^{-1}, \quad V_F = J(\phi^*V_E)J^{-1}. \quad (180)$$

- (ii) *There is a surjective real unitary $U : H_{\mathfrak{X}} \rightarrow H_{\mathfrak{Y}}$ satisfying*

$$T_{\mathfrak{Y},t}^{(2)} = U^{\odot 2} T_{\mathfrak{X},t}^{(2)}, \quad T_{\mathfrak{Y},t}^{(4)} = U^{\odot 4} T_{\mathfrak{X},t}^{(4)}. \quad (181)$$

- (iii) *For every closed loop-free 2/4-network Γ ,*

$$\mathcal{N}_{\Gamma,t}(\mathfrak{X}) = \mathcal{N}_{\Gamma,t}(\mathfrak{Y}). \quad (182)$$

Moreover, whenever a unitary U in (ii) is prescribed, the pair (ϕ, J) in (i) is obtained from that same unitary and satisfies

$$(Us)(y) = J_y s(\phi(y)) \quad \text{for a.e. } y \in N. \quad (183)$$

Disconnected networks may be omitted in (iii), since their values are products of the values of their connected components.

Proof. Assume (i) and define $(Us)(y) = J_y s(\phi(y))$. The map U is a real L^2 -unitary. Naturality of the connection Laplacian and the potential identity in (180) give $U\mathcal{L}_{\mathfrak{X}} = \mathcal{L}_{\mathfrak{Y}}U$ and hence $UA_{\mathfrak{X},t} = A_{\mathfrak{Y},t}U$. Because ϕ preserves volume and J preserves the fiber metric, the definitions (173)–(174) then give (181). Thus (i) implies (ii).

Every complete contraction is invariant under a simultaneous unitary change of all tensor legs, so (ii) implies (iii).

Conversely, the numbers in (iii) are exactly the complete loop-free contraction register of the finite tensor packet

$$(0, 0, T_{\mathfrak{X},t}^{(2)}, 0, T_{\mathfrak{X},t}^{(4)}), \quad (184)$$

and similarly for $\mathfrak{Y}$. The support-Gram, finite-dimensional orbit-separation, and compact inverse-limit argument used in the proof of [Theorem 3.11](#) therefore gives a single unitary between the active supports which transports both tensors. The quadratic flattenings are $e^{-t\mathcal{L}_{\mathfrak{X}}}$ and $e^{-t\mathcal{L}_{\mathfrak{Y}}}$. They are injective with dense range, so the active supports are the full spaces $H_{\mathfrak{X}}$ and $H_{\mathfrak{Y}}$. This proves (ii).

Finally, start with a prescribed U satisfying (ii). The proof of [Theorem 7.2](#) and [Corollary 7.3](#) applies from (156) onward: the quartic tensor makes U a vector-valued L^4 -isometry on a common dense core, [Lemma 7.1](#) gives the measurable form (183), elliptic regularity makes (ϕ, J) smooth, and comparison of the principal and first-order symbols recovers respectively g , ∇^E , and then V_E . This proves (i) for the same U and completes all implications. $\square$

The random field formulation is now an immediate, but useful, consequence. Let $(\mathbb{W}_{\mathfrak{X},x})_{x \in \mathbb{R}}$ be the fixed canvas over $H_{\mathfrak{X}}$ and set

$$\mathcal{Z}_{\mathfrak{X},t}(x) := I_2^{\mathbb{W}_{\mathfrak{X},x}}(T_{\mathfrak{X},t}^{(2)}) + I_4^{\mathbb{W}_{\mathfrak{X},x}}(T_{\mathfrak{X},t}^{(4)}), \quad x \in \mathbb{R}. \quad (185)$$

Corollary 8.5 (Geometry, networks, and one universal field). *For the two data sets in [Theorem 8.4](#), the equivalent conditions (i)–(iii) of that theorem are also equivalent to*

$$\text{Law}_{C(\mathbb{R})}(\mathcal{Z}_{\mathfrak{X},t}) = \text{Law}_{C(\mathbb{R})}(\mathcal{Z}_{\mathfrak{Y},t}). \quad (186)$$

Thus one and the same predetermined scalar Gaussian canvas is a complete scalar realization of all the closed radial heat networks, and both are complete invariants of the smooth Laplace-type geometry up to isometry and orthogonal gauge.

Proof. By [Theorem 3.11](#), equality in (186) is equivalent to a unitary transporting the two tensors on their active supports. The quadratic flattening has dense range, so those supports are the full L^2 spaces. The conclusion follows from [Theorem 8.4](#). $\square$

8.3 The exact stabilizer

Let $\text{Aut}(\mathfrak{X})$ be the group of pairs $(\psi, \mathcal{J})$ in which $\psi \in \text{Isom}(M, g)$ and $\mathcal{J} : E \rightarrow E$ is a smooth orthogonal bundle automorphism covering ψ which preserves ∇^E and V_E . Its natural action on $H_{\mathfrak{X}}$ is

$$(\rho(\psi, \mathcal{J})s)(y) := \mathcal{J}_{\psi^{-1}(y)}s(\psi^{-1}(y)). \quad (187)$$

Define the tensor stabilizer at time t by

$$\text{Stab}_t(\mathfrak{X}) := \left\{ U \in O(H_{\mathfrak{X}}) : \begin{array}{l} U^{\odot 2} T_{\mathfrak{X},t}^{(2)} = T_{\mathfrak{X},t}^{(2)}, \\ U^{\odot 4} T_{\mathfrak{X},t}^{(4)} = T_{\mathfrak{X},t}^{(4)} \end{array} \right\}. \quad (188)$$

Corollary 8.6 (Exact geometric automorphism group). *For every datum $\mathfrak{X}$ in (171), the action (187) is a group isomorphism*

$$\rho : \text{Aut}(\mathfrak{X}) \xrightarrow{\cong} \text{Stab}_t(\mathfrak{X}). \quad (189)$$

In particular, for the scalar packet $(f_{M,t}^{(2)}, f_{M,t}^{(4)})$ on a closed connected manifold,

$$\text{Stab}(f_{M,t}^{(2)}, f_{M,t}^{(4)}) \cong \{\pm 1\} \times \text{Isom}(M, g). \quad (190)$$

Proof. Naturality shows that the image of ρ is contained in the tensor stabilizer. Conversely, apply the final assertion of [Theorem 8.4](#) with $\mathfrak{Y} = \mathfrak{X}$ to any $U \in \text{Stab}_t(\mathfrak{X})$. It represents that same U by a pair (ϕ, J) with $J_y : E_{\phi(y)} \rightarrow E_y$. To match the convention in (187), put $\psi = \phi^{-1}$ and $J_x = J_{\phi^{-1}(x)} : E_x \rightarrow E_{\psi(x)}$; its induced operator is exactly U . This proves surjectivity. If two geometric automorphisms induce the same operator on all L^2 sections, their base maps and fiber maps agree (first almost everywhere, then everywhere by smoothness); hence ρ is injective.

For the trivial real line bundle, a smooth orthogonal gauge transformation is a map from connected M to $O(1) = \{\pm 1\}$ and is therefore constant. Both tensor degrees are even, so both constants occur and commute with pullback by every isometry. This gives (190). $\square$

9 Positive orthogonal Tannaka duality for the master heat packet

The preceding results single out one canonical tensor object. For a marked heat geometry $\mathfrak{X}$ and $t > 0$, set

$$\mathcal{T}_t(\mathfrak{X}) := (H_{\mathfrak{X}}, Q_{\mathfrak{X},t}, C_{\mathfrak{X},t}) := (H_{\mathfrak{X}}, T_{\mathfrak{X},t}^{(2)}, T_{\mathfrak{X},t}^{(4)}). \quad (191)$$

The flattening of $Q_{\mathfrak{X},t}$ is $e^{-t\mathcal{L}_{\mathfrak{X}}}$, while the diagonal of $C_{\mathfrak{X},t}$ is the fourth power of the radial L^4 norm on the heat range. Thus the two degrees have different intrinsic roles: Q_t recovers the generator and C_t forces the transporting unitary to be spatial. This is why (191), rather than any of its homogeneous presentations below, is the *master heat packet*.

9.1 Finite cuts and the positive pro-character

Put $H = H_{\mathfrak{X}}$, $T_2 = Q_{\mathfrak{X},t}$, and $T_4 = C_{\mathfrak{X},t}$. The intrinsic support Gram

$$D = L_{T_2} L_{T_2}^* + L_{T_4} L_{T_4}^* \quad (192)$$

is positive, trace class, and injective. For every regular threshold $\delta > 0$, meaning $\delta \notin \sigma(D)$, set

$$P_{\delta} = \mathbf{1}_{[\delta, \infty)}(D), \quad H_{\delta} = P_{\delta} H, \quad B_{\delta} = P_{\delta} D P_{\delta}, \quad T_{j,\delta} = P_{\delta}^{\otimes j} T_j \quad (j = 2, 4). \quad (193)$$

The cut space H_{δ} is finite-dimensional and the marker B_{δ} canonically records its position in the spectral flag.

Let $\mathbf{D}_{\text{free}}$ be the free real linear rigid symmetric monoidal $*$ -category generated by one self-dual object $\mathfrak{h}$, a self-adjoint endomorphism $\mathfrak{b}$, and symmetric arrows $\mathfrak{t}_j : \mathbf{1} \rightarrow \mathfrak{h}^{\otimes j}$ for $j = 2, 4$. Evaluation at $(H_{\delta}, B_{\delta}, T_{2,\delta}, T_{4,\delta})$ gives a symmetric monoidal $*$ -functor ω_{δ}^0 to finite-dimensional real Hilbert spaces. Define

$$\mathbf{D}_{\delta}^{\text{eff}} := \text{Kar}(\mathbf{D}_{\text{free}} / \ker \omega_{\delta}^0), \quad \omega_{\delta} : \mathbf{D}_{\delta}^{\text{eff}} \longrightarrow \text{Hilb}_{\text{fd}}^{\mathbb{R}}. \quad (194)$$

The induced functor is faithful, rigid, orthogonal, and symmetric monoidal. Evaluation of closed diagrams defines a multiplicative positive character

$$\chi_{\delta}(d \sqcup d') = \chi_{\delta}(d) \chi_{\delta}(d'), \quad \chi_{\delta}(\text{cl}(a^* a)) = \|\omega_{\delta}(a)\|_{\text{HS}}^2 \geq 0, \quad (195)$$

where cl denotes categorical closure by the chosen duality. The quotient removes all dimension-dependent and packet-dependent tensor relations, and the Karoubi envelope records all orthogonal summands generated by idempotents.

Proposition 9.1 (The positive cut character is complete). *For every regular δ , the character χ_δ determines the orthogonal orbit of*

$$(H_\delta, B_\delta, T_{2,\delta}, T_{4,\delta}) \quad (196)$$

and therefore determines $(D_\delta^{\text{eff}}, \omega_\delta)$ up to orthogonal symmetric monoidal equivalence. Conversely, such an equivalence preserves χ_δ .

Proof. The values of χ_δ are exactly the complete contractions, with arbitrary repetitions, of the marker and the two tensors in (196). By Proposition 3.5, these values determine the tuple up to one element of $O(H_\delta)$. Conjugating by that orthogonal map intertwines the evaluation functors, their tensor $*$ -ideals of zero arrows, and their Karoubi envelopes. Equivalently, the positive pairings

$$\langle a, b \rangle_{\chi_\delta} := \chi_\delta(\text{cl}(a^*b)) = \langle \omega_\delta^0(a), \omega_\delta^0(b) \rangle_{\text{HS}}$$

recover the null ideal and all Hilbert structures on generated morphism spaces. The converse follows from functoriality of categorical closure. $\square$

By definition, $\text{Aut}^\otimes(\omega_\delta)$ is the group of unitary symmetric monoidal natural automorphisms of ω_δ . Every such automorphism is determined by its component on $\mathfrak{h}$, and

$$G_\delta := \text{Aut}^\otimes(\omega_\delta) = \left\{ U \in O(H_\delta) : UB_\delta U^* = B_\delta, U^{\otimes j} T_{j,\delta} = T_{j,\delta}, j = 2, 4 \right\}. \quad (197)$$

If $0 < \varepsilon < \delta$ are regular, H_δ is a spectral subspace of B_ε and restriction gives a homomorphism $G_\varepsilon \rightarrow G_\delta$.

Definition 9.2 (Positive orthogonal pro-character). *The cofinal system*

$$\chi_{\mathfrak{X}, t, \bullet} := (H_\delta, D_\delta^{\text{eff}}, \omega_\delta, \chi_\delta, G_\varepsilon \rightarrow G_\delta)_\delta \quad (198)$$

is the positive orthogonal pro-character of $(\mathfrak{X}, t)$. Two pro-characters are isomorphic if, after passage to cofinal families of regular thresholds, there are compatible orthogonal isomorphisms of cut spaces transporting $B_\delta, T_{2,\delta}, T_{4,\delta}$; equivalently, there are compatible orthogonal symmetric monoidal equivalences of their effective cut categories intertwining the fiber functors and characters.

Proposition 9.3 (The inverse system reconstructs the full stabilizer). *Restriction to the spectral flag is a topological group isomorphism*

$$\text{Stab}_{O(H)}(T_2, T_4) \xrightarrow{\cong} \varprojlim_{\delta \downarrow 0} G_\delta, \quad (199)$$

where the left side carries the strong operator topology and the right side the projective-limit topology.

Proof. If U stabilizes T_2 and T_4 , then it stabilizes the support Gram D , commutes with every spectral projection P_δ , and restricts to an element of G_δ . These restrictions are compatible and determine U because $\bigcup_\delta H_\delta$ is dense.

Conversely, let $(U_\delta)_\delta$ be a compatible family. The formula $Ux = U_\delta x$ for $x \in H_\delta$ is well defined on the dense union of the cuts and is isometric. The compatible inverse family $(U_\delta^{-1})_\delta$ shows that its extension is a surjective orthogonal operator on H . For $j = 2, 4$,

$$U^{\otimes j} P_\delta^{\otimes j} T_j = P_\delta^{\otimes j} T_j.$$

Since $P_\delta^{\otimes j} T_j \rightarrow T_j$ in Hilbert tensor norm, passage to the limit gives $U^{\otimes j} T_j = T_j$. The constructions in both directions are continuous for the stated topologies, proving the isomorphism. $\square$

We write $\text{Aut}^\otimes(\omega_{\mathfrak{X}, t, \bullet})$ for the projective-limit group in (199). This is the orthogonal Tannaka group naturally carried by the separable Hilbert object through its complete cofinal family of finite rigid cuts [Saa72, DR89, Del02, EGNO15].

9.2 Full faithfulness and scalar duality

Let Heat_t be the groupoid of marked heat geometries, whose arrows are isometries together with orthogonal bundle maps preserving the connection and potential. Let $\text{Ten}_{2,4}$ be the groupoid of master packets (191), with simultaneous tensor-transporting unitaries as arrows.

Theorem 9.4 (Positive scalar Tannaka duality). *The master heat-packet functor*

$$\mathcal{T}_t : \text{Heat}_t \longrightarrow \text{Ten}_{2,4} \quad (200)$$

is fully faithful. For every pair $\mathfrak{X}, \mathfrak{Y}$, it induces the bijection

$$\text{Hom}_{\text{Heat}_t}(\mathfrak{X}, \mathfrak{Y}) \simeq \left\{ U : H_{\mathfrak{X}} \rightarrow H_{\mathfrak{Y}} : \begin{array}{l} U \text{ is a surjective real unitary,} \\ U^{\otimes j} T_{\mathfrak{X},t}^{(j)} = T_{\mathfrak{Y},t}^{(j)}, \quad j = 2, 4 \end{array} \right\}. \quad (201)$$

On isomorphism classes, the following are equivalent complete invariants:

$$[\mathfrak{X}]_{\text{isometry+gauge}}, \quad [\chi_{\mathfrak{X},t,\bullet}]_{\otimes}, \quad \text{Law}_{C(\mathbb{R})}(\mathcal{Z}_{\mathfrak{X},t}). \quad (202)$$

Moreover,

$$\text{Aut}(\mathfrak{X}) \simeq \text{Stab}_{O(H_{\mathfrak{X}})}(Q_{\mathfrak{X},t}, C_{\mathfrak{X},t}) \simeq \text{Aut}^{\otimes}(\omega_{\mathfrak{X},t,\bullet}). \quad (203)$$

Proof. The prescribed-transport statement of Theorem 8.4 identifies every simultaneous transport of (Q_t, C_t) with one unique geometric arrow, proving full faithfulness. By Proposition 9.1, the cut characters are equivalent to the finite compressed tensor orbits. An isomorphism of pro-characters gives compatible cut transports, and Proposition 9.3 extends them to one global transport of the master packets. The converse follows by restricting a master-packet transport to the common spectral flag. Finally, Corollary 8.5 identifies the closed network register with the universal scalar law, and Corollary 8.6 and Proposition 9.3 identifies all three automorphism groups. $\square$

The scalar law therefore reconstructs the complete transport torsor and all stabilizers of the master packet. The positive pro-character is its finite-cut Tannaka realization, while the homogeneous appendix gives pure-chaos carriers of the same fully faithful geometric object.

10 Conclusion: one scalar law reconstructs heat geometry

The universal Gaussian canvas turns every finite symmetric tensor packet on a separable real Hilbert space into one smooth scalar finite-chaos process whose law is a complete invariant of the packet's simultaneous orthogonal orbit. The graph-character decoder extracts the loop-free closed tensor character directly from the law; the intrinsic support Gram, its finite spectral flag, and orthogonal invariant theory recover the full orbit and every transport on the active Hilbert space.

For heat geometry, the canonical object is the positive radial packet

$$\mathcal{T}_t(\mathfrak{X}) = (H_{\mathfrak{X}}, Q_{\mathfrak{X},t}, C_{\mathfrak{X},t}).$$

Its quadratic component is the marked heat operator, while its quartic component is the first positive radial tensor carrying non-Hilbertian spatial geometry in every orthogonal fiber rank. Their simultaneous transport reconstructs the measure algebra, smooth structure, metric, bundle, connection, and self-adjoint potential. The resulting heat-packet functor is fully faithful: geometric arrows, tensor transports, and scalar laws form equivalent groupoids.

The same architecture reconstructs compact finite-dimensional RCD spaces from their Cheeger heat flows. At the categorical level, the positive cut characters recover the finite

orthogonal tensor objects, and their inverse system reconstructs the complete geometric automorphism group. The universal scalar law is therefore a positive orthogonal Tannaka realization of heat geometry, not merely an invariant of isomorphism classes.

The appendices show that the master packet admits faithful carriers in every pure Wiener chaos of order $q \geq 3$, identify the exact scale sensitivity of inverse reconstruction, and give finite statistical experiments separating arbitrary fixed candidate geometries. Together the results establish the hierarchy

$$\boxed{\begin{array}{l} \text{heat-geometric groupoid} \iff \text{positive radial } (2+4) \text{ master packet} \\ \iff \text{positive orthogonal pro-character} \iff \text{one universal scalar process law.} \end{array}} \quad (204)$$

A Homogeneous chaos carriers of the master packet

The canonical datum constructed above is the quadratic–quartic heat packet: its quadratic component is the heat operator and its quartic component is the radial L^4 tensor. This appendix proves that the positive master packet admits faithful *derived presentations* in every homogeneous Wiener chaos of order $q \geq 3$. Hence every pure chaos order at least three can carry the complete heat geometry, while the quadratic–quartic packet remains the intrinsic geometric source of all these presentations.

We first state the precise analytic class on which the constructions are defined. Throughout, Sym denotes the orthogonal symmetrizer. If $T \in H^{\odot 2}$, we identify T with the self-adjoint Hilbert–Schmidt operator A_T determined by

$$\langle A_T x, y \rangle_H = \langle T, x \odot y \rangle_{H^{\odot 2}}. \quad (205)$$

For $C \in H^{\odot 4}$, its $2+2$ flattening is the self-adjoint Hilbert–Schmidt operator on $H^{\odot 2}$ given by

$$\langle \widehat{C} \xi, \eta \rangle_{H^{\odot 2}} = \langle C, \xi \odot \eta \rangle_{H^{\odot 4}}. \quad (206)$$

The map $C \mapsto \widehat{C}$ is injective on $H^{\odot 4}$. Indeed, the tensors $\xi \odot \eta$ with $\xi, \eta \in H^{\odot 2}$ span a dense subspace of $H^{\odot 4}$.

Definition A.1 (Admissible positive master packet). *An admissible positive master packet is a triple*

$$\mathbf{T} = (H, Q, C), \quad Q \in H^{\odot 2}, \quad C \in H^{\odot 4}, \quad (207)$$

where H is a real separable Hilbert space and:

- (a) A_Q is positive and injective;
- (b) $\widehat{C}$ is positive and trace class on $H^{\odot 2}$.

An arrow $U : (H, Q, C) \rightarrow (H', Q', C')$ is a surjective real unitary satisfying

$$Q' = U^{\odot 2} Q, \quad C' = U^{\odot 4} C. \quad (208)$$

We denote the resulting groupoid by $\text{Pos}_{2,4}$.

Lemma A.2 (Positivity of a radial quartic). *Let $q \in H^{\odot 2}$ and suppose that its associated operator A_q is positive and trace class. Then the $2+2$ flattening of $q \odot q$ is positive and trace class. More precisely, for every $\xi \in H^{\odot 2}$,*

$$\langle \widehat{q \odot q} \xi, \xi \rangle = \frac{1}{3} \left(\langle q, \xi \rangle^2 + 2 \|A_q^{1/2} A_\xi A_q^{1/2}\|_{\mathcal{S}_2}^2 \right) \geq 0. \quad (209)$$

Moreover,

$$\text{Tr}(\widehat{q \odot q}) = \frac{1}{3} \left((\text{Tr } A_q)^2 + 2 \|A_q\|_{\mathcal{S}_2}^2 \right). \quad (210)$$

Proof. Diagonalize A_q and expand the orthogonal symmetrizer of $q \otimes q$. The three pairings of four tensor legs give (209). In an eigenbasis of A_q with eigenvalues (λ_i) , the map $X \mapsto A_q X A_q$ on self-adjoint Hilbert–Schmidt operators has eigenvalues λ_i^2 on the diagonal directions and $\lambda_i \lambda_j$ on the symmetric off-diagonal directions. Adding the rank-one term $\xi \mapsto \langle q, \xi \rangle q$ and summing these eigenvalues gives (210). $\square$

The injectivity in Definition A.1 is exactly the condition used earlier to show that the active support of the heat packet is the whole L^2 space. The trace-class hypothesis is the condition which makes the square-root factor used below Hilbert–Schmidt. Both conditions hold for the radial heat packets of this paper. Indeed, $A_Q = e^{-t\mathcal{L}}$ is positive and injective at every $t > 0$. Moreover, the pointwise radial quadratic tensor is positive and finite rank. By Lemma A.2, the $2 + 2$ flattening of its symmetric square is positive and trace class. Heat smoothing, finite fiber rank, and compactness make the integral of these positive flattenings trace class.

A.1 Anchor padding in every order $q \geq 4$

Let a be a fixed unit vector orthogonal to H . For $T \in H^{\odot k}$ and $q \geq k$, define

$$\text{Pad}_{q,k}^a(T) := \binom{q}{k} \text{Sym}(a^{\otimes(q-k)} \otimes T) \in (\mathbb{R}a \oplus H)^{\odot q}. \quad (211)$$

The normalization in (211) makes this the sum over the distinct placements of the k legs of T ; no conclusion below depends on that normalization. For $q \geq 4$, put

$$\begin{aligned} K_q(\mathbf{T}) &= \mathbb{R}a \oplus H, \\ P_q(\mathbf{T}) &= a^{\otimes q}, \\ S_q(\mathbf{T}) &= \text{Pad}_{q,2}^a(Q) + \text{Pad}_{q,4}^a(C). \end{aligned} \quad (212)$$

Both marked tensors $P_q(\mathbf{T})$ and $S_q(\mathbf{T})$ have pure degree q .

Theorem A.3 (Faithfulness of anchor padding). *Let $q \geq 4$ and let $\mathbf{T} = (H, Q, C)$ and $\mathbf{T}' = (H', Q', C')$ be admissible positive master packets. Then*

$$\begin{aligned} \exists U : H &\longrightarrow H' \text{ satisfying (208)} \\ \iff \exists W : K_q(\mathbf{T}) &\longrightarrow K_q(\mathbf{T}') \text{ unitary such that} \\ P_q(\mathbf{T}') &= W^{\odot q} P_q(\mathbf{T}), \quad S_q(\mathbf{T}') = W^{\odot q} S_q(\mathbf{T}). \end{aligned} \quad (213)$$

More precisely, in the subgroupoid whose arrows preserve the oriented anchor, $Wa = a'$, the functor

$$\mathcal{E}_q(U) = \text{Id}_{\mathbb{R}a} \oplus U \quad (214)$$

is fully faithful and identifies $\text{Pos}_{2,4}$ with the subgroupoid formed by the packets (212) and the anchor-preserving arrows. If the anchor is remembered only through P_q , the same assertion holds without qualification for odd q . For even q , the kernel on arrows is exactly the auxiliary reflection $(-\text{Id}_{\mathbb{R}a}) \oplus \text{Id}_H$; quotienting by this order-two kernel again gives an equivalent groupoid.

Proof. If U transports Q and C , then $W = \text{Id}_{\mathbb{R}a} \oplus U$ transports both tensors in (212). This proves the reverse implication in (213) and defines (214).

Conversely, suppose that W transports P_q . Equality of two unit rank-one symmetric tensors gives

$$Wa = \varepsilon a', \quad \varepsilon \in \{+1, -1\}, \quad \varepsilon^q = 1. \quad (215)$$

Hence $W(H) = H'$; write $U = W|_H$. Relative to the orthogonal splitting

$$(\mathbb{R}a \oplus H)^{\odot q} = \bigoplus_{k=0}^q a^{\odot(q-k)} \odot H^{\odot k}, \quad (216)$$

the two summands of S_q have different H -degrees, namely 2 and 4. Projecting the equality $S_q(\mathbf{T}') = W^{\odot q} S_q(\mathbf{T})$ onto these two summands gives

$$Q' = \varepsilon^{q-2} U^{\odot 2} Q, \quad C' = \varepsilon^{q-4} U^{\odot 4} C. \quad (217)$$

If q is odd, (215) forces $\varepsilon = 1$. If q is even, both exponents in (217) are even. Thus in all cases U satisfies (208).

If $Wa = a'$, the restriction U determines W uniquely, which proves full faithfulness for oriented anchors. Without the orientation, the only two possible extensions of a fixed U are obtained by the two signs in (215); the negative sign occurs exactly when q is even. This proves the final assertion. $\square$

The jointly active support of the marked pair (P_q, S_q) is all of $K_q(\mathbf{T})$. Indeed, P_q supplies the anchor line. Contracting the $q-2$ anchor legs of S_q kills the quartic summand and recovers a nonzero multiple of Q . Since A_Q is injective and self-adjoint, its range is dense in H , so no direction of H is invisible.

A.2 The minimal positive cubic dilation

Padding cannot place a quartic tensor directly in degree three. The positive flattening of C provides the missing factorization. Set

$$E_H = H^{\odot 2}, \quad L_C = \overline{\text{Ran } \widehat{C}^{1/2}} \subset E_H, \quad B_C : E_H \longrightarrow L_C, \quad B_C \xi = \widehat{C}^{1/2} \xi. \quad (218)$$

Then B_C is Hilbert–Schmidt and has dense range, and

$$B_C^* B_C = \widehat{C}, \quad \|B_C\|_{\mathcal{S}_2}^2 = \text{Tr}(\widehat{C}). \quad (219)$$

This factorization is minimal in the usual Kolmogorov sense.

Lemma A.4 (Uniqueness of the minimal positive factor). *Suppose that $\widetilde{B} : E_H \rightarrow \widetilde{L}$ is Hilbert–Schmidt, $\overline{\text{Ran } \widetilde{B}} = \widetilde{L}$, and $\widetilde{B}^* \widetilde{B} = \widehat{C}$. There is a unique surjective unitary $J : L_C \rightarrow \widetilde{L}$ such that*

$$\widetilde{B} = J B_C. \quad (220)$$

Proof. On $\text{Ran } B_C$, define $J(B_C \xi) = \widetilde{B} \xi$. This is well defined and isometric because

$$\langle B_C \xi, B_C \eta \rangle = \langle \xi, \widehat{C} \eta \rangle = \langle \widetilde{B} \xi, \widetilde{B} \eta \rangle.$$

The two ranges are dense, so J extends uniquely to a surjective unitary. $\square$

Let $\beta_{B_C} \in L_C \otimes H^{\odot 2}$ be the Hilbert tensor corresponding to the Hilbert–Schmidt map B_C :

$$\langle \beta_{B_C}, \ell \otimes \xi \rangle = \langle B_C \xi, \ell \rangle_{L_C}, \quad \ell \in L_C, \quad \xi \in H^{\odot 2}. \quad (221)$$

Identify the positive trace-class operator $B_C B_C^*$ on L_C with its quadratic tensor. On

$$K_3(\mathbf{T}) = \mathbb{R}a \oplus H \oplus L_C \quad (222)$$

define

$$\begin{aligned} R_{\mathbf{T}} &= Q \oplus (-B_C B_C^*) \in (H \oplus L_C)^{\odot 2}, \\ P_3(\mathbf{T}) &= a^{\otimes 3}, \\ S_3(\mathbf{T}) &= 3 \operatorname{Sym}(a \otimes R_{\mathbf{T}}) + 3 \operatorname{Sym}(\beta_{B_C}) \in K_3(\mathbf{T})^{\odot 3}. \end{aligned} \quad (223)$$

Here β_{B_C} is first placed in $L_C \otimes H \otimes H$ and then symmetrized. The factors 3 make the two terms the sums over the three possible positions of the distinguished leg.

Theorem A.5 (Full faithfulness of the positive cubic dilation). *The assignment*

$$\mathcal{E}_3 : (H, Q, C) \longmapsto (K_3(\mathbf{T}), P_3(\mathbf{T}), S_3(\mathbf{T})) \quad (224)$$

is fully faithful from $\operatorname{Pos}_{2,4}$ to the groupoid of marked symmetric cubic pairs. Explicitly,

$$\begin{aligned} &\left\{ U : H \rightarrow H' : Q' = U^{\odot 2} Q, C' = U^{\odot 4} C \right\} \\ &\xrightarrow{\cong} \left\{ W : K_3(\mathbf{T}) \rightarrow K_3(\mathbf{T}') : \begin{array}{l} P_3(\mathbf{T}') = W^{\odot 3} P_3(\mathbf{T}), \\ S_3(\mathbf{T}') = W^{\odot 3} S_3(\mathbf{T}) \end{array} \right\}. \end{aligned} \quad (225)$$

For a master arrow U , the corresponding cubic arrow is

$$\tilde{U} = \operatorname{Id}_{\mathbb{R}a} \oplus U \oplus U^{\odot 2} \Big|_{L_C}. \quad (226)$$

Consequently the cubic pair and the quadratic–quartic packet have equivalent orthogonal orbits and isomorphic stabilizer groupoids.

Proof. Assume first that U is a master arrow and put $U_2 = U^{\odot 2}$. Equivariance of the $2 + 2$ flattening gives

$$\hat{C}' = U_2 \hat{C} U_2^*. \quad (227)$$

Functional calculus then gives $\hat{C}'^{1/2} = U_2 \hat{C}^{1/2} U_2^*$ and $U_2 L_C = L_{C'}$. Therefore

$$V_U := U_2|_{L_C} : L_C \longrightarrow L_{C'} \quad \text{is unitary,} \quad B_{C'} U_2 = V_U B_C. \quad (228)$$

It follows from (221) that

$$\beta_{B_{C'}} = (V_U \otimes U^{\odot 2}) \beta_{B_C}, \quad B_{C'} B_{C'}^* = V_U B_C B_C^* V_U^*. \quad (229)$$

Thus the unitary in (226) transports both cubic tensors.

Conversely, let W be a cubic arrow. Since the exponent is odd,

$$W^{\odot 3} a^{\otimes 3} = a'^{\otimes 3} \implies W a = a'. \quad (230)$$

Hence W restricts to a unitary $W_0 : H \oplus L_C \rightarrow H' \oplus L_{C'}$. Contracting one leg of S_3 against a gives exactly

$$\iota_a S_3 = R_{\mathbf{T}}. \quad (231)$$

Consequently W_0 conjugates the self-adjoint operator

$$A_{R_{\mathbf{T}}} = A_Q \oplus (-B_C B_C^*) \quad (232)$$

to $A_{R_{\mathbf{T}'}}$. The positive spectral subspace of $A_{R_{\mathbf{T}}}$ is precisely H : A_Q is positive and injective. Its negative spectral subspace is precisely L_C : the dense-range property of B_C implies $\ker(B_C B_C^*) = \ker B_C^* = \{0\}$. Spectral projections are preserved by unitary conjugacy, and therefore

$$W_0 = U \oplus V \quad (233)$$

for unitaries $U : H \rightarrow H'$ and $V : L_C \rightarrow L_{C'}$. The positive block in (232) yields $Q' = U^{\odot 2}Q$.

Subtract $3\text{Sym}(a \otimes R)$ from S_3 and project one tensor leg to L_C and the other two to H . By (221), the remaining transport identity is

$$\beta_{B_{C'}} = (V \otimes U^{\odot 2})\beta_{B_C}, \quad \text{equivalently} \quad B_{C'}U^{\odot 2} = VB_C. \quad (234)$$

Taking adjoints and products gives

$$\hat{C}' = B_{C'}^* B_{C'} = U^{\odot 2} \hat{C} (U^{\odot 2})^*. \quad (235)$$

Injectivity of the flattening map now gives $C' = U^{\odot 4}C$. Hence U is a master arrow. Finally, (234) determines V uniquely on the dense range of B_C , and then on all of L_C . Thus W is exactly the unitary in (226), proving the bijection on every morphism set. $\square$

The active support of (P_3, S_3) is all of $K_3(\mathbf{T})$. The first tensor supplies a . Equation (231) and the injectivity of both diagonal blocks in (232) show that the tensor legs of S_3 have dense support in $H \oplus L_C$. This observation will allow us to apply Replica–Tannaka without an unobservable orthogonal complement.

A.3 One marked scalar law in a pure homogeneous chaos

We record the typed form of the universal-canvas theorem. It is needed because a homogeneous presentation consists of two cubic, or two q -ic, tensors rather than one tensor in each of two different grades.

Lemma A.6 (Quadratic–exponential independence). *Let $(a_j, b_j)_{1 \leq j \leq N}$ be distinct pairs in $[0, \infty) \times \mathbb{R}$. Then the functions*

$$t \mapsto e^{-a_j t^2 + b_j t}, \quad 1 \leq j \leq N, \quad (236)$$

are linearly independent on $\mathbb{R}$.

Proof. Assume $\sum_j c_j e^{-a_j t^2 + b_j t} = 0$. Among indices with $c_j \neq 0$, choose a_0 minimal and, among those with $a_j = a_0$, choose b_0 maximal. Multiplication by $e^{a_0 t^2 - b_0 t}$ and passage to $t \rightarrow +\infty$ leaves only the coefficient attached to (a_0, b_0) , which must vanish. Iteration proves that every coefficient is zero. $\square$

Lemma A.7 (Typed homogeneous Canvas–Replica reconstruction). *Fix $q \geq 1$ and distinct real tags $s_1, \dots, s_r$. Let $\mathbf{f} = (f_1, \dots, f_r)$ with $f_\alpha \in H^{\odot q}$, and let $(\mathbb{W}_u)_{u \in \mathbb{R}}$ be the fixed Gaussian canvas with*

$$\mathbb{E}[\mathbb{W}_u(h)\mathbb{W}_v(k)] = e^{-(u-v)^2} \langle h, k \rangle_H. \quad (237)$$

Define the real process

$$\mathcal{Z}_{\mathbf{f}}^{(q)}(u) := \sum_{\alpha=1}^r e^{s_\alpha u} I_q^{\mathbb{W}_u}(f_\alpha). \quad (238)$$

Then every $\mathcal{Z}_{\mathbf{f}}^{(q)}(u)$ belongs to the pure q -th Wiener chaos, and the process has a continuous version. The typed process is generally nonstationary, while its driving Gaussian canvas remains stationary. For a second family $\mathbf{f}'$ with the same tags,

$$\begin{aligned} \text{Law}_{C(\mathbb{R})}(\mathcal{Z}_{\mathbf{f}}^{(q)}) &= \text{Law}_{C(\mathbb{R})}(\mathcal{Z}_{\mathbf{f}'}^{(q)}) \\ \iff \exists U : S(\mathbf{f}) &\rightarrow S(\mathbf{f}') \text{ unitary such that } f'_\alpha = U^{\odot q} f_\alpha \text{ for every } \alpha. \end{aligned} \quad (239)$$

Here $S(\mathbf{f})$ is the closed joint active support of the family.

Proof. The reverse implication follows from equivariance of multiple Wiener integrals. Equality of process laws gives equality of every finite mixed moment, since finite-chaos random variables have moments of all orders; no converse moment-determinacy assertion is used. For the direct implication, expand every mixed moment of (238) by Wick's rule. A loop-free q -regular multigraph with adjacency multiplicities (m_{ij}) and a vertex typing τ contributes its typed contraction multiplied by

$$\exp \left(\sum_i s_{\tau(i)} u_i - \sum_{i < j} m_{ij} (u_i - u_j)^2 \right). \quad (240)$$

For each fixed moment order choose a vector v outside the finite union of the hyperplanes and quadrics on which two distinct pairs

$$\left(\sum_{i < j} m_{ij} (v_i - v_j)^2, \sum_i s_{\tau(i)} v_i \right) \quad (241)$$

coincide. On setting $u_i = tv_i$, the coefficients become distinct functions e^{-at^2+bt} . Their linear independence is Lemma A.6; hence the process law determines every loop-free contraction with every vertex type retained.

Define the colored support Gram

$$D_{\mathbf{f}} = \sum_{\alpha=1}^r L_{f_{\alpha}} L_{f_{\alpha}}^*.$$

The typed loop-free register determines its spectral moments and all polynomially opened contractions on every finite spectral cap. Applying Proposition 3.5 to the colored tuple $(P_{\delta}^{\otimes q} f_{\alpha})_{\alpha}$ together with the marker $P_{\delta} D_{\mathbf{f}} P_{\delta}$ produces compatible orthogonal transports on all caps. The compact inverse-limit construction of Theorem 3.2 then yields one common unitary transporting every f_{α} . Continuity follows from the smooth universal canvas and hypercontractivity. More explicitly, in the single-isonormal realization of (58), the deterministic kernel is the Hilbert-smooth curve

$$u \mapsto \sum_{\alpha=1}^r e^{s_{\alpha} u} (\Psi(u)^{\otimes q} \otimes f_{\alpha}) \in (\mathcal{G} \otimes H)^{\odot q},$$

whose derivatives are bounded on compact intervals. $\square$

Apply the lemma with two tags, for example $s_P = 0$ and $s_S = 1$. For $q \geq 4$ use the pair in (212); for $q = 3$ use the pair in (223). Thus, with the appropriate ambient Hilbert space $K_q(\mathbf{T})$, set

$$\boxed{\mathcal{Z}_{\mathbf{T}}^{(q)}(u) = I_q^{\mathbb{W}u}(P_q(\mathbf{T})) + e^u I_q^{\mathbb{W}u}(S_q(\mathbf{T})), \quad q \geq 3.} \quad (242)$$

Although it contains two deterministically tagged tensor types, every random variable in (242) lies entirely in one homogeneous chaos.

Corollary A.8 (Homogeneous presentations of the master packet). *Let $\mathbf{T}$ and $\mathbf{T}'$ be admissible positive master packets. For every $q \geq 3$,*

$$\begin{aligned} \text{Law}_{C(\mathbb{R})}(\mathcal{Z}_{\mathbf{T}}^{(q)}) &= \text{Law}_{C(\mathbb{R})}(\mathcal{Z}_{\mathbf{T}'}^{(q)}) \\ \iff \exists U : H \rightarrow H' \text{ satisfying } Q' &= U^{\odot 2} Q, \quad C' = U^{\odot 4} C. \end{aligned} \quad (243)$$

For the radial heat packets of Theorem 8.4, these conditions are further equivalent to equivalence of the underlying manifold, metric, Euclidean bundle, metric connection, and self-adjoint potential.

Proof. The two marked families are jointly active by the observations following [Theorems A.3](#) and [A.5](#). The typed Canvas–Replica equivalence [\(239\)](#), followed respectively by [Theorem A.3](#) or [Theorem A.5](#), proves [\(243\)](#). The geometric assertion is exactly the deterministic master equivalence already proved in [Theorem 8.4](#). $\square$

Remark A.9 (The master datum and its homogeneous carriers). *Corollary A.8 establishes the exact chain*

$$\boxed{\text{pure tagged chaos of order } q \geq 3 \iff \text{Orb}(Q, C) \iff \text{heat geometry.}} \quad (244)$$

For $q \geq 4$, the carrier separates the quadratic and quartic tensors by occupation number relative to a canonical anchor line. For $q = 3$, it uses the Kolmogorov-minimal positive factor of $\hat{C}$, canonical up to a unique unitary by [Lemma A.4](#). Hence every order $q \geq 3$ supports a faithful homogeneous realization, while (Q, C) remains the intrinsic geometric packet from which all carriers are functorially generated.

B Sharpness and inverse stability

The reconstruction theorems above are exact. It is natural to ask whether they admit a uniform inverse modulus on geometrically bounded classes. At a fixed positive heat time the answer is sharply negative under the usual zeroth-order curvature bounds. The following example also explains why a quantitative version of Replica–Tannaka reconstruction must retain information at vanishing heat scales.

B.1 An oscillating gauge field asymptotically invisible at every fixed scale

Let $M = \mathbb{R}^2 / (2\pi\mathbb{Z})^2$ be the flat two-torus, let $E = M \times \mathbb{R}^2$, and put

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

For an integer $k \geq 1$, consider the metric connection

$$\nabla^{(k)} = d + a_k(x)J dy, \quad a_k(x) = \frac{\sin(kx)}{k}, \quad (245)$$

and let $\nabla^{(0)} = d$. Its curvature is

$$F_{\nabla^{(k)}} = \cos(kx)J dx \wedge dy. \quad (246)$$

Thus

$$\|F_{\nabla^{(k)}}\|_{L^\infty} = c_J, \quad \int_M |F_{\nabla^{(k)}}|^2 dv = \frac{1}{2} \text{Vol}(M) c_J^2, \quad (247)$$

where $c_J := |J| > 0$ for the fixed norm on $\mathfrak{so}(2)$, and for every k . In particular, no orthogonal gauge transformation can make these curvatures converge uniformly to the curvature of the flat connection.

For a fixed marked time $t > 0$, let $Q_{k,t}$ and $C_{k,t}$ be the quadratic and radial quartic tensors of [\(141\)](#), constructed from $\nabla^{(k)}$, and define on the fixed universal canvas

$$\mathcal{Y}_{k,t}(u) = I_2^{\mathbb{W}_u}(Q_{k,t}) + I_4^{\mathbb{W}_u}(C_{k,t}), \quad u \in \mathbb{R}. \quad (248)$$

Equip $C(\mathbb{R})$, with its locally uniform topology, with the metric

$$\rho_{\text{loc}}(z, z') = \sum_{j \geq 1} 2^{-j} \min(1, \|z - z'\|_{C([-j, j])}). \quad (249)$$

Below, d_{BL} denotes the bounded-Lipschitz metric associated with ρ_{loc} : the supremum of the differences of integrals over all functions bounded by one and 1-Lipschitz for ρ_{loc} .

Proposition B.1 (Fixed-time replica data do not control curvature). *For every fixed $t > 0$,*

$$Q_{k,t} \longrightarrow Q_{0,t} \quad \text{in } H^{\odot 2}, \quad C_{k,t} \longrightarrow C_{0,t} \quad \text{in } H^{\odot 4}, \quad (250)$$

where $H = L^2(M, \mathbb{R}^2)$. Consequently,

$$\text{Law}_{C(\mathbb{R})}(\mathcal{Y}_{k,t}) \longrightarrow \text{Law}_{C(\mathbb{R})}(\mathcal{Y}_{0,t}) \quad (251)$$

weakly, although $\|F_{\nabla^{(k)}}\|_{L^\infty} = c_J$ and the limiting connection is flat. More strongly, for every $p > 2$ and $R > 0$, under the common canvas coupling,

$$\mathbb{E} \|\mathcal{Y}_{k,t} - \mathcal{Y}_{0,t}\|_{C([-R,R])}^p \longrightarrow 0. \quad (252)$$

For every $0 < \tau < T < \infty$,

$$\sup_{t \in [\tau, T]} d_{\text{BL}}(\text{Law}_{C(\mathbb{R})}(\mathcal{Y}_{k,t}), \text{Law}_{C(\mathbb{R})}(\mathcal{Y}_{0,t})) \longrightarrow 0. \quad (253)$$

Proof. In the standard trivialization, the Bochner Laplacian is

$$L_k = (\nabla^{(k)})^* \nabla^{(k)} = L_0 - 2a_k J \partial_y + a_k^2 \text{Id}, \quad L_0 = -\partial_x^2 - \partial_y^2. \quad (254)$$

Thus, for $s > 0$,

$$\|(L_k - L_0)e^{-sL_0}\|_{L^2 \rightarrow L^2} \leq \frac{C}{k\sqrt{s}} + \frac{1}{k^2}. \quad (255)$$

The heat semigroups of the metric connections are dominated in norm by the scalar heat semigroup. In dimension two this gives, uniformly in k ,

$$\|e^{-sL_k}\|_{L^2 \rightarrow L^\infty} \leq C_s, \quad C_s \leq C(1 + s^{-1/2}) \quad (0 < s \leq 1). \quad (256)$$

For $T > 0$, Duhamel's formula, first on the common core $C^\infty(M, \mathbb{R}^2)$ and then by closure, therefore yields

$$\begin{aligned} \|e^{-TL_k} - e^{-TL_0}\|_{L^2 \rightarrow L^\infty} &\leq \int_0^T \|e^{-(T-s)L_k}\|_{L^2 \rightarrow L^\infty} \|(L_k - L_0)e^{-sL_0}\|_{L^2 \rightarrow L^2} ds \\ &\leq \frac{C_T}{k}. \end{aligned} \quad (257)$$

The integrability at the endpoints follows from $(T-s)^{-1/2}s^{-1/2} \in L^1(0, T)$.

Take $T = t/2$. The $L^2 \rightarrow L^\infty$ norm first controls the essential supremum of the row-operator norms. The heat kernels are smooth and the volume has full support, so the bound holds at every x . Evaluation at x turns e^{-TL_k} into a map $K_{k,t,x} : H \rightarrow \mathbb{R}^2$. Since the target has dimension two, (257) implies, uniformly in x ,

$$\|K_{k,t,x} - K_{0,t,x}\|_{\text{HS}} \leq \frac{C_t}{k}, \quad \sup_k \sup_x \|K_{k,t,x}\|_{\text{HS}} < \infty. \quad (258)$$

For $q_{k,t,x} = \sum_{\alpha=1}^2 K_{k,t,x}^* e_\alpha \otimes K_{k,t,x}^* e_\alpha$, the elementary tensor estimate gives

$$\sup_x \|q_{k,t,x} - q_{0,t,x}\|_{H^{\otimes 2}} \leq \frac{C_t}{k}. \quad (259)$$

Integration over the fixed compact base proves convergence of the quadratic tensors. Applying $\|a \odot a - b \odot b\| \leq (\|a\| + \|b\|) \|a - b\|$ before integration proves convergence of the quartic tensors.

Put $f_{k,2} = Q_{k,t}$ and $f_{k,4} = C_{k,t}$. Realize all the processes in (248) with the same canvas isonormal family. The multiple-integral isometry gives

$$\mathbb{E} \left| I_q^{\mathbb{W}^u}(f_{k,q} - f_{0,q}) \right|^2 = q! \|f_{k,q} - f_{0,q}\|^2, \quad q \in \{2, 4\}. \quad (260)$$

Applying Lemma 3.8 to the difference packet $(0, 0, f_{k,2} - f_{0,2}, 0, f_{k,4} - f_{0,4})$ proves (252), and therefore (251).

The constants in (255)–(259) are uniform for $t \in [\tau, T]$. The quantitative bound (53), applied on each $[-j, j]$, therefore gives $\sup_{t \in [\tau, T]} \mathbb{E} \rho_{\text{loc}}(\mathcal{Y}_{k,t}, \mathcal{Y}_{0,t}) \rightarrow 0$ by dominated convergence of the series in (249). The coupling inequality for the bounded-Lipschitz metric therefore implies (253). Finally, (247) proves the claimed geometric nonconvergence. $\square$

Corollary B.2 (No zeroth-order inverse modulus). *Let d_{law} be any metric inducing weak convergence on probability measures on $C(\mathbb{R})$. There is no function $\omega(s) \downarrow 0$ as $s \downarrow 0$ such that, on the class of connections (245),*

$$\|F_{\nabla}\|_{L^\infty} \leq \omega(d_{\text{law}}(\text{Law}(\mathcal{Y}_{\nabla,t}), \text{Law}(\mathcal{Y}_{d,t}))). \quad (261)$$

This remains false if the datum contains the whole curve of positive heat times with the compact-open topology induced by d_{BL} on every interval $[\tau, T] \Subset (0, \infty)$.

Proof. Apply the proposed estimate to $\nabla^{(k)}$. Its left-hand side is c_J by (247), while its right-hand side tends to zero by Proposition B.1. $\square$

B.2 Inverse continuity on compact geometric classes

We first make the ambient topology explicit.

Definition B.3 (Smooth Cheeger–Gromov–gauge convergence). *Let $\mathfrak{X}_j = (M_j, g_j, E_j, \nabla^j, V_j)$ and $\mathfrak{X} = (M, g, E, \nabla, V)$ be smooth heat geometries modulo isometry and orthogonal gauge. We say that $[\mathfrak{X}_j] \rightarrow [\mathfrak{X}]$ smoothly if, for all sufficiently large j , the base dimensions and bundle ranks agree and there exist diffeomorphisms $\phi_j : M \rightarrow M_j$ and smooth orthogonal bundle isomorphisms $J_j : E \rightarrow \phi_j^* E_j$ such that*

$$\phi_j^* g_j \longrightarrow g, \quad J_j^{-1} \phi_j^* \nabla^j J_j \longrightarrow \nabla, \quad J_j^{-1} \phi_j^* V_j J_j \longrightarrow V \quad (262)$$

in every C^m norm on the fixed compact base M . The resulting quotient topology on isometry–gauge classes is called the smooth Cheeger–Gromov–gauge topology. In the statement below $\mathcal{K}$ is assumed compact and metrizable in this topology.

Although low-order quantitative stability fails, exact reconstruction immediately yields a sharp qualitative statement on genuinely compact families. Let $\mathcal{K}$ be a compact metrizable subset of this smooth moduli space. Fix $t > 0$, and let

$$\mathcal{S}_t : \mathcal{K} \longrightarrow \mathcal{P}(C(\mathbb{R})) \quad (263)$$

send a gauge geometry to the law of its universal-canvas process.

Proposition B.4 (Compactness–uniqueness stability). *The map $\mathcal{S}_t$ is a homeomorphism from $\mathcal{K}$ onto its compact image. In particular, after choosing any compatible metrics $d_{\mathcal{K}}$ and d_{law} , there is a nondecreasing modulus $\omega_{\mathcal{K},t}$, with $\omega_{\mathcal{K},t}(s) \rightarrow 0$ as $s \downarrow 0$, such that*

$$d_{\mathcal{K}}(X, Y) \leq \omega_{\mathcal{K},t}(d_{\text{law}}(\mathcal{S}_t X, \mathcal{S}_t Y)) \quad (X, Y \in \mathcal{K}). \quad (264)$$

Proof. Convergence in the sense of [Definition B.3](#), after the defining diffeomorphism and gauge identifications, gives convergence of the Laplace-type coefficients in every C^m norm. Standard parabolic regularity then gives smooth convergence of the heat kernels at the marked positive time. The quadratic and radial quartic tensors converge in Hilbert tensor norm, so [Equation \(53\)](#) implies weak convergence of the continuous canvas laws. Thus $\mathcal{S}_t$ is continuous. It is injective by the gauge reconstruction theorem with potential and the universal-canvas theorem. A continuous injection from a compact space into the Hausdorff space $\mathcal{P}(C(\mathbb{R}))$ is a homeomorphism onto its image. Uniform continuity of its inverse gives [\(264\)](#). $\square$

Corollary B.5 (Minimum-distance consistency on compact geometric classes). *Let $\mathcal{X}$, $\mathcal{S}_t$, $d_{\mathcal{X}}$, and d_{law} be as in [Proposition B.4](#). Fix $X_0 \in \mathcal{X}$, let $Z_1, Z_2, \dots$ be independent $C(\mathbb{R})$ -valued observations with common law $\mathcal{S}_t X_0$, and put*

$$\hat{\nu}_n = \frac{1}{n} \sum_{j=1}^n \delta_{Z_j}. \quad (265)$$

Suppose $\hat{X}_n \in \mathcal{X}$ satisfies

$$d_{\text{law}}(\hat{\nu}_n, \mathcal{S}_t \hat{X}_n) \leq \inf_{X \in \mathcal{X}} d_{\text{law}}(\hat{\nu}_n, \mathcal{S}_t X) + \frac{1}{n}. \quad (266)$$

Then

$$d_{\mathcal{X}}(\hat{X}_n, X_0) \longrightarrow 0 \quad \text{almost surely.} \quad (267)$$

Proof. The path space $C(\mathbb{R})$ with the compact-open topology is Polish, so its empirical measures converge weakly almost surely to their common law. Since d_{law} metrizes weak convergence,

$$d_{\text{law}}(\hat{\nu}_n, \mathcal{S}_t X_0) \longrightarrow 0 \quad \text{almost surely.}$$

By [\(266\)](#) and the triangle inequality,

$$d_{\text{law}}(\mathcal{S}_t \hat{X}_n, \mathcal{S}_t X_0) \leq 2d_{\text{law}}(\hat{\nu}_n, \mathcal{S}_t X_0) + \frac{1}{n}.$$

The right-hand side tends to zero. Continuity of the inverse of $\mathcal{S}_t$ on its compact image, equivalently [\(264\)](#), yields [\(267\)](#). $\square$

Remark B.6 (Scale-sensitive quantitative reconstruction). *The oscillatory family pinpoints the analytic mechanism behind inverse conditioning: high-frequency curvature is encoded at progressively shorter heat scales. A quantitative Replica–Tannaka theory is therefore naturally multiscale. It combines short-time renormalization with an approximate spatialization theorem selecting one nearly common gauge near singular orthogonal orbits. On compact smooth moduli classes, [Proposition B.4](#) already gives the exact topological form of this inverse control.*

C Finite statistical identification

The preceding corollary is an exact identifiability statement. It also gives a finite statistical experiment for every fixed pair of models. For $B > 0$, write

$$c_B(y) = \max\{-B, \min\{B, y\}\}. \quad (268)$$

Theorem C.1 (Finite-coordinate exponentially consistent test). *Let $\mathbf{f}$ and $\mathbf{f}'$ be two finite tensor packets which are not orthogonally equivalent on their minimal active supports. Then*

there exist an integer $r \geq 1$, a rational configuration $x = (x_1, \dots, x_r) \in E_r \cap \mathbb{Q}^r$, and $B > 0$ such that

$$m = \mathbb{E} c_B \left(\prod_{a=1}^r \mathcal{Z}_{\mathbf{f}}(x_a) \right), \quad m' = \mathbb{E} c_B \left(\prod_{a=1}^r \mathcal{Z}_{\mathbf{f}'}(x_a) \right) \quad (269)$$

are different. Put $\delta = |m - m'| > 0$.

Suppose that $Z^{(1)}, \dots, Z^{(n)}$ are independent observed paths, all drawn either from $\text{Law}(\mathcal{Z}_{\mathbf{f}})$ or from $\text{Law}(\mathcal{Z}_{\mathbf{f}'})$, and set

$$\hat{m}_n = \frac{1}{n} \sum_{j=1}^n c_B \left(\prod_{a=1}^r Z^{(j)}(x_a) \right). \quad (270)$$

Choose the first model when $|\hat{m}_n - m| \leq |\hat{m}_n - m'|$ and the second otherwise. Then

$$\max \{ \mathbb{P}_{\mathbf{f}}(\text{error}), \mathbb{P}_{\mathbf{f}'}(\text{error}) \} \leq 2 \exp \left(-\frac{n\delta^2}{8B^2} \right). \quad (271)$$

In particular, the test is strongly consistent and uses only finitely many rational evaluations of each path.

Proof. By the proof of [Corollary 3.14](#), there are r and a rational $x \in E_r$ for which the untruncated mixed moments

$$\mathbb{E} \prod_{a=1}^r \mathcal{Z}_{\mathbf{f}}(x_a) \quad \text{and} \quad \mathbb{E} \prod_{a=1}^r \mathcal{Z}_{\mathbf{f}'}(x_a)$$

differ. Products of finite-chaos variables are integrable, because every finite-chaos variable belongs to every finite L^p . Hence the expectations of the clipped products converge to the two displayed moments as $B \rightarrow \infty$. They therefore remain different for some finite B .

Under either hypothesis, the summands in (270) are independent and lie in $[-B, B]$. A wrong nearest-mean decision requires a deviation of at least $\delta/2$ from the correct mean. Hoeffding's inequality [[Hoe63](#)] gives

$$\mathbb{P} \left(|\hat{m}_n - \mathbb{E} \hat{m}_n| \geq \frac{\delta}{2} \right) \leq 2 \exp \left(-\frac{n\delta^2}{8B^2} \right),$$

which proves (271). The almost-sure consistency also follows directly from the strong law. $\square$

Corollary C.2 (Finite candidate classes). *Let $L \geq 2$ and let $\mathbf{f}^{(1)}, \dots, \mathbf{f}^{(L)}$ be pairwise inequivalent finite tensor packets. There exist $P \leq \binom{L}{2}$ bounded continuous statistics*

$$\psi_1, \dots, \psi_P : C(\mathbb{R}) \longrightarrow [-1, 1], \quad (272)$$

each depending only on finitely many rational evaluations, such that the mean vectors

$$v_i = (\mathbb{E} \psi_1(\mathcal{Z}_{\mathbf{f}^{(i)}}), \dots, \mathbb{E} \psi_P(\mathcal{Z}_{\mathbf{f}^{(i)}})) \in [-1, 1]^P \quad (273)$$

are pairwise distinct. Let $d_ = \min_{i \neq j} \|v_i - v_j\|_\infty > 0$. From n independent paths define the empirical feature vector $\hat{v}_n$ and choose a nearest vector among $v_1, \dots, v_L$ in the ℓ^∞ norm. Under model i ,*

$$\mathbb{P}_i(\text{misclassification}) \leq 2P \exp \left(-\frac{nd_*^2}{8} \right). \quad (274)$$

Proof. For each unordered pair (i, j) , apply [Theorem C.1](#) and divide its clipped product statistic by the corresponding B . Collecting these statistics gives [\(272\)](#); the coordinate attached to (i, j) separates v_i from v_j . If every coordinate of $\hat{v}_n - v_i$ has absolute value less than $d_*/2$, then v_i is the unique nearest model vector. Hoeffding’s inequality for variables in $[-1, 1]$, followed by a union bound over the P coordinates, gives [\(274\)](#). $\square$

Remark C.3 (Pairwise and finite-class identification). *The separating replica order, rational configuration, clipping level, and exponential gap are determined by the candidate laws. The theorem therefore gives a finite experiment for every fixed pair and a single finite feature family for every finite candidate class. On compact geometric model classes, [Proposition B.4](#) upgrades exact injectivity to inverse continuity and yields consistent minimum-distance reconstruction.*

AI-assisted document audit

A large language model was used solely as a document-auditing tool. The audit identified minor errors in the proof of [Proposition 6.2](#) (*RCD heat flows satisfy the reconstruction hypotheses*) and directed the author’s attention to existing references for the heat-kernel bounds, L^p -analyticity, and intrinsic-metric rigidity used there.

The audit also identified an intermediate local computation that had been left implicit in the proof of [Theorem 7.2](#) (*A scalar replica observable hears an orthogonal gauge field*). More precisely, after writing $\nabla^F = \tilde{\nabla} + B$, the local expansion of the difference of the two Bochner Laplacians was checked in normal coordinates and a $\tilde{\nabla}$ -normal frame. This yields the first-order term displayed in [\(164\)](#), namely

$$-2 \sum_{j=1}^{\dim N} B(X_j) \partial_{X_j},$$

from which equality of the two differential operators forces $B = 0$.

Finally, the audit detected theorem-type cross-reference errors, including references to [Lemma 7.1](#) (*Simultaneous L^2 – L^4 rigidity*) and [Proposition 9.3](#) (*The inverse system reconstructs the full stabilizer*). These typographical errors were corrected without any change to the mathematical content.

These corrections concern an auxiliary RCD verification, one local Bochner-Laplacian computation, and typographical cross-references. They do not alter the constructions, statements of the principal theorems, or logical architecture of the paper. All mathematical statements and citations were independently verified by the author.

References

- [AGS14a] Luigi Ambrosio, Nicola Gigli, and Giuseppe Savaré. Calculus and heat flow in metric measure spaces and applications to spaces with Ricci bounds from below. *Invent. Math.*, 195(2):289–391, 2014.
- [AGS14b] Luigi Ambrosio, Nicola Gigli, and Giuseppe Savaré. Metric measure spaces with Riemannian Ricci curvature bounded from below. *Duke Math. J.*, 163(7):1405–1490, 2014.
- [ABtE12] Wolfgang Arendt, Markus Biegert, and A. F. M. ter Elst. Diffusion determines the manifold. *J. Reine Angew. Math.*, 667:1–25, 2012.
- [Ass26a] Obayda Julien Assaad. Finite Gaussian reconstruction of polynomial orbits: from correlated moments to oscillatory periods. Preprint, arXiv:2608.14475, 2026.
- [Ass26b] Obayda Julien Assaad. Weak limits of Wiener chaos: primitive–Fock classification and Hilbert–Stein extraction. Preprint, arXiv:2608.12492, 2026.

- [BGL14] Dominique Bakry, Ivan Gentil, and Michel Ledoux. *Analysis and Geometry of Markov Diffusion Operators*. Grundlehren der mathematischen Wissenschaften 348. Springer, 2014.
- [Bra37] Richard Brauer. On algebras which are connected with the semisimple continuous groups. *Ann. of Math. (2)*, 38(4):857–872, 1937.
- [CL25] Mihajlo Cekić and Thibault Lefeuvre. Isospectral connections, ergodicity of frame flows, and polynomial maps between spheres. *Ann. Sci. Éc. Norm. Supér. (4)*, 58(1):203–229, 2025.
- [Dan22] Nguyen Viet Dang. Gaussian free fields and Riemannian rigidity. *Probab. Math. Phys.*, 3(1):1–34, 2022.
- [Del02] Pierre Deligne. Catégories tensorielles. *Mosc. Math. J.*, 2(2):227–248, 2002.
- [DR89] Sergio Doplicher and John E. Roberts. A new duality theory for compact groups. *Invent. Math.*, 98(1):157–218, 1989.
- [EKS15] Matthias Erbar, Kazumasa Kuwada, and Karl-Theodor Sturm. On the equivalence of the entropic curvature-dimension condition and Bochner’s inequality on metric measure spaces. *Invent. Math.*, 201(3):993–1071, 2015.
- [EGNO15] Pavel Etingof, Shlomo Gelaki, Dmitri Nikshych, and Victor Ostrik. *Tensor Categories*. Mathematical Surveys and Monographs 205. American Mathematical Society, Providence, RI, 2015.
- [FLS07] Michael H. Freedman, László Lovász, and Alexander Schrijver. Reflection positivity, rank connectivity, and homomorphisms of graphs. *J. Amer. Math. Soc.*, 20(1):37–51, 2007.
- [FOT11] Masatoshi Fukushima, Yoichi Oshima, and Masayoshi Takeda. *Dirichlet Forms and Symmetric Markov Processes*. De Gruyter Studies in Mathematics 19. De Gruyter, second revised and extended edition, 2011.
- [Gig18] Nicola Gigli. Nonsmooth differential geometry—an approach tailored for spaces with Ricci curvature bounded from below. *Mem. Amer. Math. Soc.*, 251(1196), 2018.
- [GWW92] Carolyn Gordon, David Webb, and Scott Wolpert. Isospectral plane domains and surfaces via Riemannian orbifolds. *Invent. Math.*, 110(1):1–22, 1992.
- [GJ87] Peter Greim and James E. Jamison. Hilbert spaces have the strong Banach–Stone property for Bochner spaces. *Math. Z.*, 196(4):511–515, 1987.
- [HMP24] Ronan Herry, Dominique Malicet, and Guillaume Poly. Limit distributions for polynomials with independent and identically distributed entries. Preprint, arXiv:2412.06749, 2024.
- [Hoe63] Wassily Hoeffding. Probability inequalities for sums of bounded random variables. *J. Amer. Statist. Assoc.*, 58(301):13–30, 1963.
- [Iss18] Leon Isserlis. On a formula for the product-moment coefficient of any order of a normal frequency distribution in any number of variables. *Biometrika*, 12(1/2):134–139, 1918.
- [Jan97] Svante Janson. *Gaussian Hilbert Spaces*, volume 129 of *Cambridge Tracts in Mathematics*. Cambridge University Press, Cambridge, 1997.
- [JLZ16] Renjin Jiang, Huaqian Li, and Huichun Zhang. Heat kernel bounds on metric measure spaces and some applications. *Potential Anal.*, 44(3):601–627, 2016.
- [Kac66] Mark Kac. Can one hear the shape of a drum? *Amer. Math. Monthly*, 73(4, part 2):1–23, 1966.
- [KOP18] Yaroslav Kurylev, Lauri Oksanen, and Gabriel P. Paternain. Inverse problems for the connection Laplacian. *J. Differential Geom.*, 110(3):457–494, 2018.
- [Lam58] John Lamperti. On the isometries of certain function-spaces. *Pacific J. Math.*, 8(3):459–466, 1958.
- [LSW18] Daniel Lenz, Marcel Schmidt, and Melchior Wirth. Geometric properties of Dirichlet forms under order isomorphisms. Preprint, arXiv:1801.08326, 2018.

- [LS09] László Lovász and Balázs Szegedy. Contractors and connectors of graph algebras. *J. Graph Theory*, 60(1):11–30, 2009.
- [Mil64] John Milnor. Eigenvalues of the Laplace operator on certain manifolds. *Proc. Natl. Acad. Sci. USA*, 51(4):542, 1964.
- [PT11] Giovanni Peccati and Murad S. Taqqu. *Wiener Chaos: Moments, Cumulants and Diagrams*. Bocconi & Springer Series 1. Springer, Milan, 2011.
- [Saa72] Neantro Saavedra Rivano. *Catégories tannakiennes*. Lecture Notes in Mathematics 265. Springer, Berlin, 1972.
- [Sch24] Joe Schaefer. Triple products of eigenfunctions and spectral geometry. Preprint, arXiv:2406.12868, 2024.
- [Sch08] Alexander Schrijver. Tensor subalgebras and first fundamental theorems in invariant theory. *J. Algebra*, 319(3):1305–1319, 2008.
- [Sch38] Isaac J. Schoenberg. Metric spaces and positive definite functions. *Trans. Amer. Math. Soc.*, 44(3):522–536, 1938.
- [Sou78] A. R. Sourour. The isometries of $L^p(\Omega, X)$. *J. Funct. Anal.*, 30(2):276–285, 1978.
- [Sun85] Toshikazu Sunada. Riemannian coverings and isospectral manifolds. *Ann. of Math. (2)*, 121(1):169–186, 1985.
- [Vig80] Marie-France Vignéras. Variétés riemanniennes isospectrales et non isométriques. *Ann. of Math. (2)*, 112(1):21–32, 1980.
- [Wey46] Hermann Weyl. *The Classical Groups: Their Invariants and Representations*. Princeton University Press, Princeton, NJ, second edition, 1946.
- [WWX26] Xing Wang, Emmett L. Wyman, and Yakun Xi. Hearing the shape of a drum by knocking around. *Bull. Lond. Math. Soc.*, 58(1):e70255, 2026.
- [You25] Ben Young. The converse of the real orthogonal Holant theorem. In *52nd International Colloquium on Automata, Languages, and Programming (ICALP 2025)*, volume 334 of *Leibniz International Proceedings in Informatics (LIPIcs)*, pages 136:1–136:20. Schloss Dagstuhl–Leibniz-Zentrum für Informatik, 2025.